

# CBOE - Research Report

Cboe Global Markets, Inc. · Financial Services / Financial Data & Stock Exchanges  
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INTERNAL

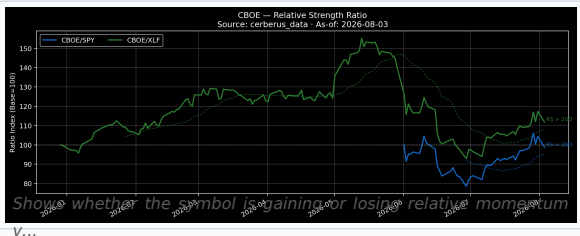
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|                                                                                                          |                                               |                               |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------|
| <b>SCORE</b><br><b>52.45/100</b>                                                                         | <b>VERDICT</b><br><b>Monitor for reclaim</b>  | <b>PRICE</b><br><b>297.83</b> |
| <b>SETUP</b><br><b>Breakout @ 311.06</b>                                                                 | <b>TARGET GAP</b><br><b>target gap +6.10%</b> | <b>R/R</b><br><b>2.0:1</b>    |
| <b>NEXT EARNINGS</b><br><b>2026-10-30</b>                                                                |                                               |                               |
| <b>Catalysts</b> Earnings on 2026-10-30   Dividend recorded 2026-05-25                                   |                                               |                               |
| <b>Risks</b> No elevated near-term risk flagged.                                                         |                                               |                               |
| <b>Setup</b> Setup: Breakout   Trigger: Level / Status   Invalid: 272.25 / EMA21 285.37   Target: 348.99 |                                               |                               |

## Executive Summary

- Score: 52.45/100
- MONITOR
- State: Hold / Neutral
- State: Uptrend / High-Vol Range

## Relative Strength Ratio



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Executive Summary & Verdict

**\*\*NEUTRAL HOLD / NEUTRAL\*\***

NOTE

Technical Score: 52.45/100 · Verdict: MONITOR · Advice: Hold / Neutral

| Pillar              | Assessment                     |
|---------------------|--------------------------------|
| Market Cycle        | Stage 2 — Uptrend OK           |
| Trend / Regime      | Uptrend / High-Vol Range       |
| TTM Squeeze         | —                              |
| Valuation (P/E)     | 22.66x                         |
| Return on Equity    | 26.30% OK                      |
| Net Margin          | 26.73% OK                      |
| Earnings Growth     | 50.20% OK                      |
| Debt/Equity         | 28.09 OK                       |
| Analyst Mean Target | \$316.00 — +6.10% from current |

Data Quality

|                                     |                                          |
|-------------------------------------|------------------------------------------|
| PROVIDER QUALITY SCORE<br>90.00/100 | EFFECTIVE REPORT CONFIDENCE<br>45.00/100 |
| FRESHNESS STATUS<br>STALE DATA      |                                          |

Validation Warnings

|                                                             |
|-------------------------------------------------------------|
| <b>WARNING</b><br>- Dividend yield looks extreme at 93.00%. |
|-------------------------------------------------------------|

Data Freshness Brief

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| STATUS<br>STALE        | FRESH / STALE / UNKNOWN<br>0 / 25 / 1                     |
| TRACKED DATASETS<br>26 | DETAIL LOCATION<br>Appendix / Data Freshness & Provenance |

Investment Case

|                                          |                    |
|------------------------------------------|--------------------|
| THESIS<br>Hold / Neutral (MONITOR setup) | VERDICT<br>MONITOR |
|------------------------------------------|--------------------|

|                                               |                                                       |
|-----------------------------------------------|-------------------------------------------------------|
| <b>TIME HORIZON</b><br>Swing                  | <b>ENTRY / TRIGGER</b><br>311.06                      |
| <b>STOP / INVALIDATION</b><br>272.25 (-8.59%) | <b>TRADE TARGET</b><br>348.99 (17.18%)                |
| <b>ANALYST MEAN TARGET</b><br>316.00 (6.10%)  | <b>REWARD / RISK</b><br>2.00:1                        |
| <b>NEXT CATALYST</b><br>2026-10-30            | <b>POSITIONING</b><br>major \$-3.32M / retail \$8.11M |

Investment Insight

| Lens        | Insight                                 |
|-------------|-----------------------------------------|
| Read        | favorable asymmetry                     |
| Implication | setup supports action if trigger clears |
| Watch Next  | trigger 311.06 and catalyst 2026-10-30  |
| Confidence  | high                                    |

Decision Dashboard

| Lens             | Read                | Lens         | Read                     |
|------------------|---------------------|--------------|--------------------------|
| Trend            | 100.00              | Momentum     | 69.22                    |
| Risk Adjusted    | 42.40               | Confidence   | 4.91                     |
| Valuation        | Fair                | Flows        | Distribution             |
| Macro Regime     | neutral             | Event Window | 87d                      |
| Relative Context | -1.19% vs benchmark | Setup        | Uptrend / High-Vol Range |

Dashboard Insight

| Lens        | Insight                                 |
|-------------|-----------------------------------------|
| Read        | signal stack weak                       |
| Implication | position sizing should stay selective   |
| Watch Next  | flows distribution and event window 87d |
| Confidence  | medium / conflicting signals            |

Report Summary

| Lens               | Summary         |
|--------------------|-----------------|
| Thesis             | Hold / Neutral  |
| Score / Verdict    | 52.45 / MONITOR |
| Strongest Evidence | Trend 100.00    |

| Lens          | Summary       |
|---------------|---------------|
| Biggest Drag  | Drawdown 0.00 |
| Catalyst      | 2026-10-30    |
| Invalidation  | 272.25        |
| Flow Backdrop | \$-3.32M      |

Summary Insight

| Lens        | Insight                                           |
|-------------|---------------------------------------------------|
| Read        | summary supports current call                     |
| Implication | thesis still needs cleaner confirmation           |
| Watch Next  | watch catalyst 2026-10-30 and flow follow-through |
| Confidence  | medium / stale inputs                             |

Thesis Delta Brief

| Lens   | Read                                                                 |
|--------|----------------------------------------------------------------------|
| Status | Unavailable                                                          |
| Reason | No previous same-profile report artifact found for delta comparison. |

Anomaly Watchlist

| Anomaly                                     | Action                                                             |
|---------------------------------------------|--------------------------------------------------------------------|
| short ratio moved materially vs 20D average | treat as exception path until it normalizes or confirms the thesis |

Anomaly Insight

| Lens        | Insight                                                           |
|-------------|-------------------------------------------------------------------|
| Read        | outlier conditions active                                         |
| Implication | tail events can dominate normal signal interpretation             |
| Watch Next  | watch whether the anomaly fades, persists, or starts moving price |
| Confidence  | medium / stale inputs                                             |

Conviction Gradecard

| Pillar        | Weight | Score  |
|---------------|--------|--------|
| Trend         | 35%    | 100.00 |
| Momentum      | 30%    | 69.22  |
| Risk-Adjusted | 20%    | 42.40  |
| Valuation     | 15%    | 55.00  |

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| Pillar    | Weight | Score                  |
|-----------|--------|------------------------|
| Composite | 100%   | 72.49 (B / Accumulate) |

Confidence Decomposition

| Driver           | Value                                                                       |
|------------------|-----------------------------------------------------------------------------|
| Strong: Trend    | 100.00                                                                      |
| Strong: Momentum | 69.22                                                                       |
| Strong: Volume   | 57.49                                                                       |
| Weak: VWAP       | 0.00                                                                        |
| Weak: Drawdown   | 0.00                                                                        |
| Weak: Bollinger  | 21.97                                                                       |
| Read             | Support led by Trend, Momentum, Volume; drag from VWAP, Drawdown, Bollinger |

NOTE

Score Bars: Momentum 69 | Trend 100 | RiskAdj 42 | VolStab 57

Price Snapshot

Snapshot

|                       |                          |
|-----------------------|--------------------------|
| PRICE<br>\$297.83     | MARKET CAP<br>\$30.39B   |
| 52W HIGH<br>\$371.18  | 52W LOW<br>\$227.15      |
| BETA<br>0.41          | DIVIDEND YIELD<br>93.00% |
| TRAILING P/E<br>22.66 | FORWARD P/E<br>19.06     |
| EPS (TTM)<br>12.84    | SHARES OUT<br>104.43M    |
| FLOAT<br>104.03M      | FLOAT %<br>99.61%        |

Returns

| 1Y      | YTD     | MTD    | WTD    | 1D     |
|---------|---------|--------|--------|--------|
| +18.72% | +20.04% | +0.00% | +0.00% | -4.00% |

Setup / Risk Dashboard

|                         |                           |
|-------------------------|---------------------------|
| CURRENT PRICE<br>297.83 | PRIMARY TRIGGER<br>311.06 |
|-------------------------|---------------------------|

STOP  
272.25

STREET HIGH TARGET  
380.00

ATR(14)  
12.79

TRADE TARGET  
348.99

REWARD / RISK  
2.00:1

EXPECTED MOVE BAND  
4.29%

Setup Insight

| Lens        | Insight                                  |
|-------------|------------------------------------------|
| Read        | risk map actionable                      |
| Implication | trade can be planned before entry        |
| Watch Next  | break 311.06 with stop held above 272.25 |
| Confidence  | high                                     |

Setup Quality Grade

GRADE  
C

COMPOSITE SETUP SCORE  
61.16

SIGNAL STACK  
trend 100.00, momentum 69.22, confidence 4.91

REWARD / RISK  
2.00:1

UPSIDE TO TARGET  
17.18%

ROOM TO STOP  
-8.59%

Catalyst Summary Card

NEXT EARNINGS  
2026-10-30

EARNINGS COUNTDOWN  
87d

EXPECTED MOVE PROXY  
4.29%

LATEST DIVIDEND / SPLIT  
2026-05-25

TOP CATALYSTS  
Earnings on 2026-10-30 ; Dividend recorded 2026-05-25 ; Analyst target gap +6.10%

TOP RISKS  
No elevated near-term risk flagged.

Scenario Analysis

| Scenario | Trigger        | Path                          | Consensus Context            | Fail Condition       | Probability |
|----------|----------------|-------------------------------|------------------------------|----------------------|-------------|
| Bull     | break > 311.06 | trade target 348.99 (+17.18%) | analyst mean 316.00 (+6.10%) | lose 272.25          | 21%         |
| Base     | hold > 297.83  | drift to 313.18 (+5.15%)      | analyst mean 316.00 (+6.10%) | fail to reach 348.99 | 45%         |
| Bear     | lose 272.25    | path 272.25 (-8.59%)          | analyst mean 316.00 (+6.10%) | reclaim 311.06       | 34%         |

Key Debate

| Side              | Framing                                              |
|-------------------|------------------------------------------------------|
| Bull Case         | trend and momentum support continuation              |
| Bear Case         | institutional flow is fading under the surface       |
| Decision Variable | confidence 4.91, major flow \$-3.32M, momentum 69.22 |

Key Reasons

| Side                   | Detail                                                            |
|------------------------|-------------------------------------------------------------------|
| Reasons To Be Long     | trend score 100                                                   |
| Reasons To Be Long     | momentum score 69                                                 |
| Reasons Not To Be Long | signal confidence 5%                                              |
| Reasons Not To Be Long | 5D major flow \$-3.32M                                            |
| Reasons Not To Be Long | short pressure rising                                             |
| What Changes The Call  | Hold / Neutral breaks if stop fails or catalyst path deteriorates |



Figure 1. Price Chart - Candlestick + SMA + Volume + RSI (1Y)

Chart Read: Shows the primary price trend, participation, and momentum backdrop.

Read: setup offers about 17.2% upside versus 9.4% stop distance. Why it matters: combines trend, participation, and execution levels in one view. Source: local market data · As-of: 2026-08-03



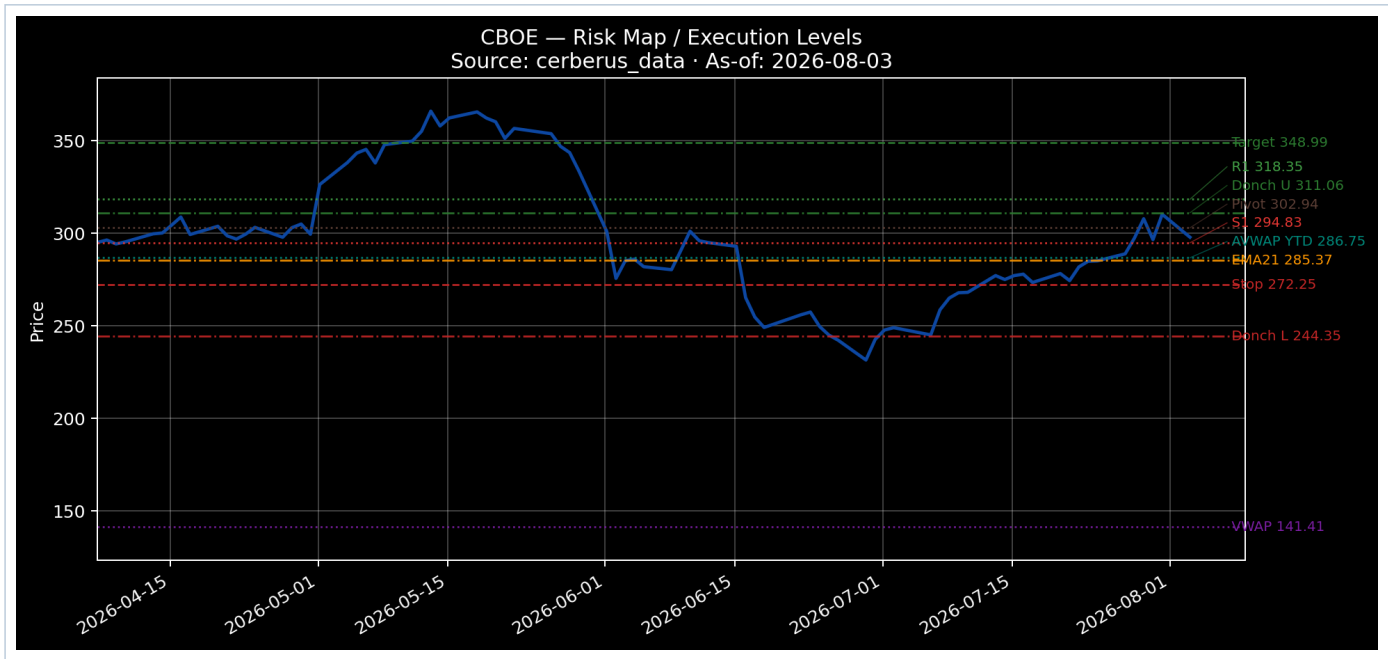


Figure 2. Risk Map and Execution Levels

Chart Read: Maps trigger, stop, and target levels into one execution view.

Read: payoff skew is favorable if trigger quality holds. Why it matters: makes trigger, invalidation, and target explicit. Source: technical indicators · As-of: 2026-08-03

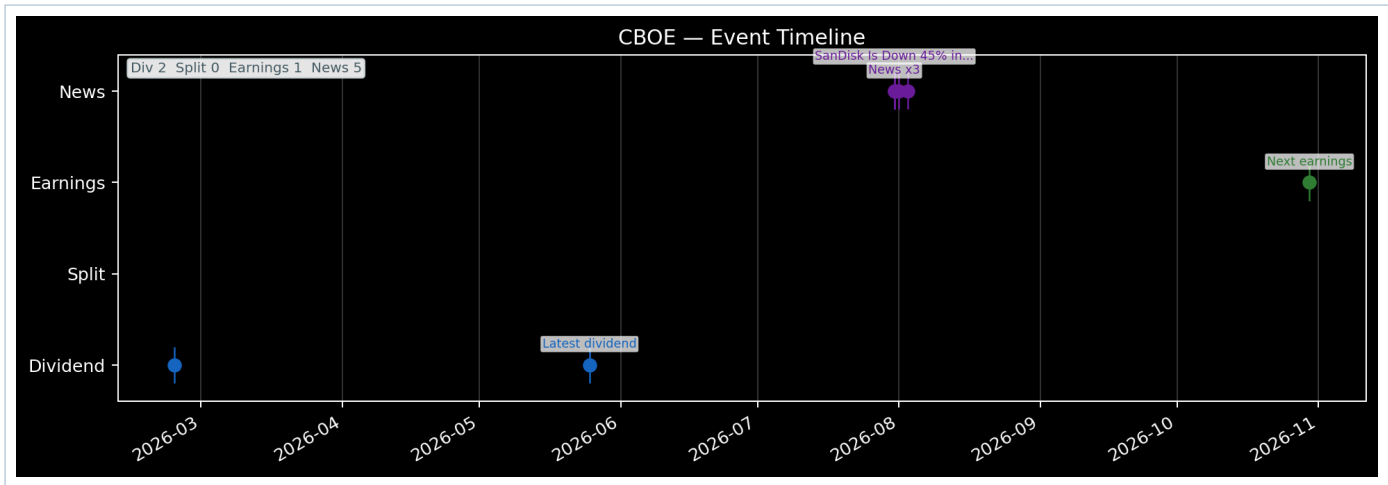


Figure 3. Catalyst and Event Timeline

Chart Read: Shows when key corporate events and headline catalysts occurred or are upcoming.

Read: next event window centers on 2026-10-30. Why it matters: timing of catalysts drives path dependence for the setup. Source: SEC events + earnings + news · As-of: 2026-07-31

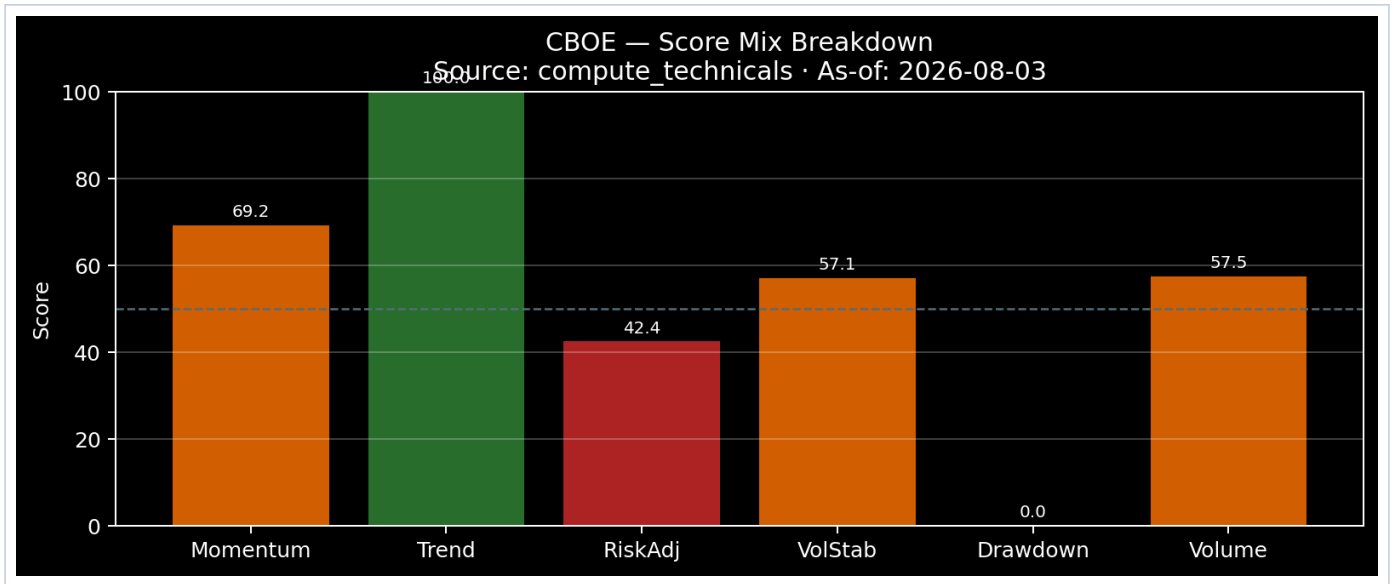


Figure 4. Score Mix Breakdown

Chart Read: Shows which technical sub-scores are carrying conviction and which remain weak.

Why it matters: shows which score drivers are carrying or dragging conviction. Source: technical scoring · As-of: 2026-08-03

Peer / Benchmark Snapshot

|                                   |                                          |
|-----------------------------------|------------------------------------------|
| PRIMARY BENCHMARK<br>SPY          | 6M RELATIVE VS PRIMARY<br>-1.19%         |
| SECTOR BENCHMARK<br>XLF           | 6M RELATIVE VS SECTOR<br>6.62%           |
| MEDIAN PEER 6M RELATIVE<br>13.78% | NET MARGIN VS PEER MEDIAN<br>26.73% vs - |

Relative Strength / Benchmark Context

|                                          |                                 |
|------------------------------------------|---------------------------------|
| 1Y RETURN (SYMBOL)<br>18.72              | YTD RETURN (SYMBOL)<br>20.04    |
| BETA VS MARKET<br>0.41                   | PRIMARY BENCHMARK<br>SPY        |
| BENCHMARK-RELATIVE RETURN (6M)<br>-1.19% | SECTOR/PEER BENCHMARK<br>XLF    |
| INDEX MEMBERSHIP<br>sp500                | EXCHANGE / TYPE<br>BTS / EQUITY |
| INTERPRETATION<br>inline                 |                                 |

| Horizon | Symbol  | Benchmark | Spread  |
|---------|---------|-----------|---------|
| 1M      | 19.62%  | 1.73%     | 17.88%  |
| 3M      | -13.24% | 0.14%     | -13.38% |

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| Horizon                                                                                                                  | Symbol | Benchmark                              | Spread |
|--------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------|--------|
| 6M                                                                                                                       | 11.36% | 0.14%                                  | 11.21% |
| 1Y                                                                                                                       | 20.49% | 0.14%                                  | 20.35% |
| PRIMARY BENCHMARK<br>SPY                                                                                                 |        | 6M RELATIVE RETURN VS SPY<br>-1.19%    |        |
| SECTOR BENCHMARK<br>XLF                                                                                                  |        | 6M RELATIVE RETURN VS SECTOR<br>6.62%  |        |
| PEER RANK<br>7/7                                                                                                         |        | PEER PERCENTILE<br>0.00%               |        |
| BREADTH (PEERS POSITIVE 1Y)<br>5/6                                                                                       |        | BREADTH (PEERS POSITIVE 6M REL)<br>6/6 |        |
| MA PARTICIPATION<br>4/4 above                                                                                            |        | INDEX MEMBERSHIP<br>sp500              |        |
| INTERPRETATION<br>lagging primary benchmark; leading sector peer benchmark; bottom-tier peer momentum; above 4/4 key MAs |        | PEER READ<br>neglected                 |        |
| LEADERSHIP<br>Laggard                                                                                                    |        | CROWDING<br>Neglected                  |        |
| POSTURE<br>Cyclical                                                                                                      |        | PRIMARY RELATIVE<br>-1.19%             |        |
| SECTOR RELATIVE<br>6.62%                                                                                                 |        | PEER PERCENTILE<br>0.00%               |        |
| CONCLUSION<br>laggard; vs primary benchmark negative; sector leadership confirmed; neglected positioning                 |        |                                        |        |

Relative Strength Insight

| Lens        | Insight                                       |
|-------------|-----------------------------------------------|
| Read        | relative leadership absent                    |
| Implication | benchmark or peers may offer cleaner exposure |
| Watch Next  | relative ratio and breadth must stay aligned  |
| Confidence  | high                                          |

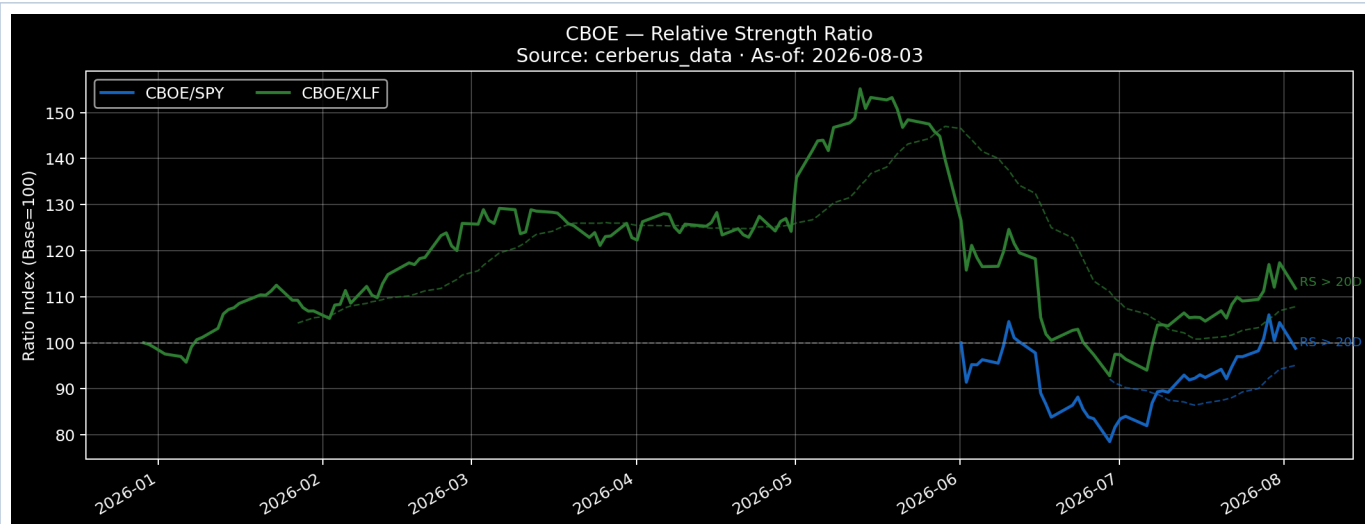


Figure 5. Relative Strength Ratio

Chart Read: Shows whether the symbol is gaining or losing relative momentum versus its benchmark.

Read: lagging SPY by 1.2%. Why it matters: confirms whether the symbol is actually outperforming its benchmark set. Source: local market data vs benchmark · As-of: 2026-08-03

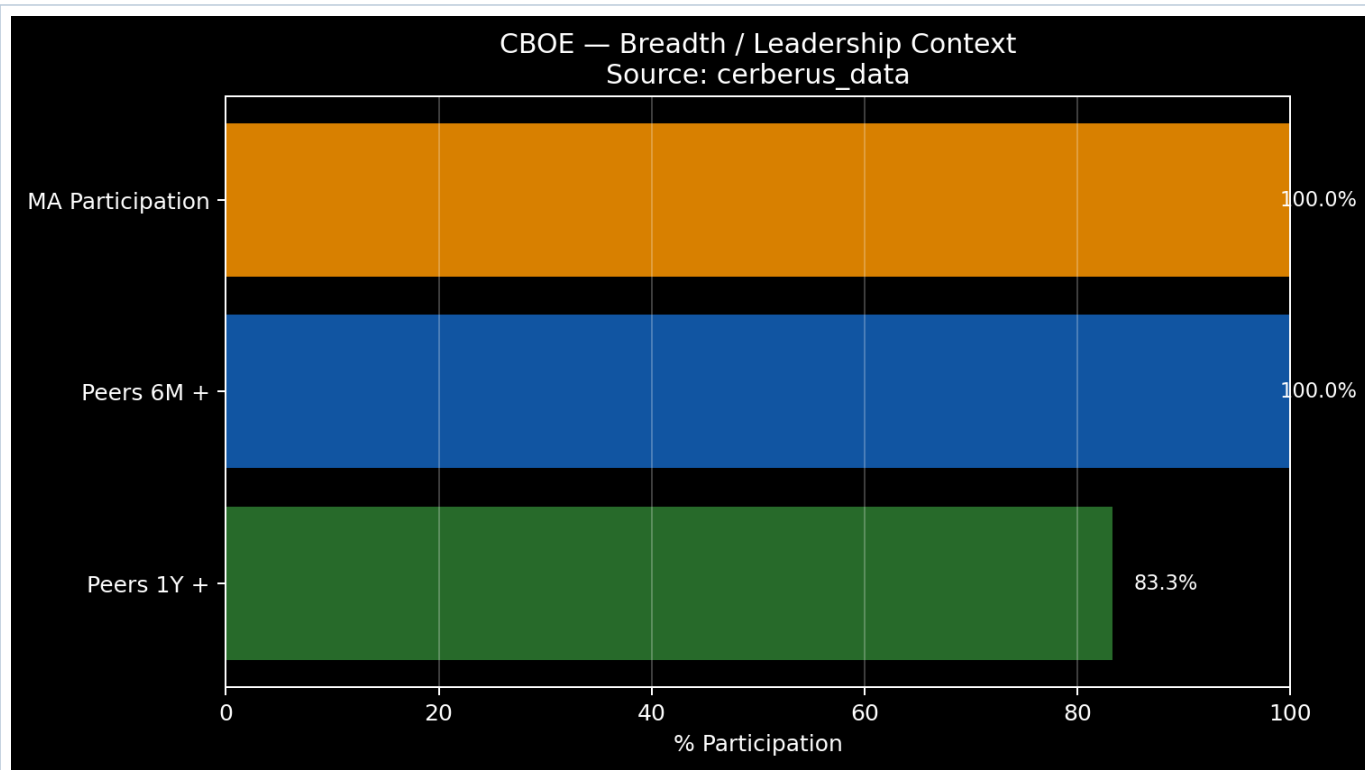


Figure 6. Breadth and Leadership Context

Chart Read: Shows how broadly supported the move is across peers and moving-average participation.

Why it matters: tests whether leadership is broad or narrow versus peers. Source: peer snapshot / local market data · As-of: 2026-08-03

## Peer Allocation View

### NOTE

Peer cut vs SPY using local index/sector mapping.

| Lens               | Subject          | Peer Medi... | Verdict     |
|--------------------|------------------|--------------|-------------|
| 6M Relative Return | -1.19%           | 13.78%       | Laggard     |
| 1Y Return          | 18.72%           | 23.08%       | Inline      |
| Net Margin         | 26.73%           | —            | Unavailable |
| Operating Margin   | 34.63%           | —            | Unavailable |
| 5D Major Flow      | \$-3.32M         | \$-5.96M     | Leader      |
| Panel Verdict      | laggard / leader | 6 peers      | SPY         |

Benchmark-relative context vs SPY using local sector/index peers.

| Symbol | 1Y Return | 6M Rel | Net Mar... | Op Margin | 5D Major Flow |
|--------|-----------|--------|------------|-----------|---------------|
| ALL    | 33.43%    | 26.63% | —          | —         | \$-8.52M      |
| AJG    | -12.11%   | 20.49% | —          | —         | \$1.26M       |
| ACGL   | 14.91%    | 13.81% | —          | —         | \$-38.00M     |
| AIZ    | 53.20%    | 13.76% | —          | —         | \$-3.41M      |
| AFL    | 31.25%    | 12.88% | —          | —         | \$-1.29M      |
| AIG    | 4.08%     | 7.84%  | —          | —         | \$-26.89M     |

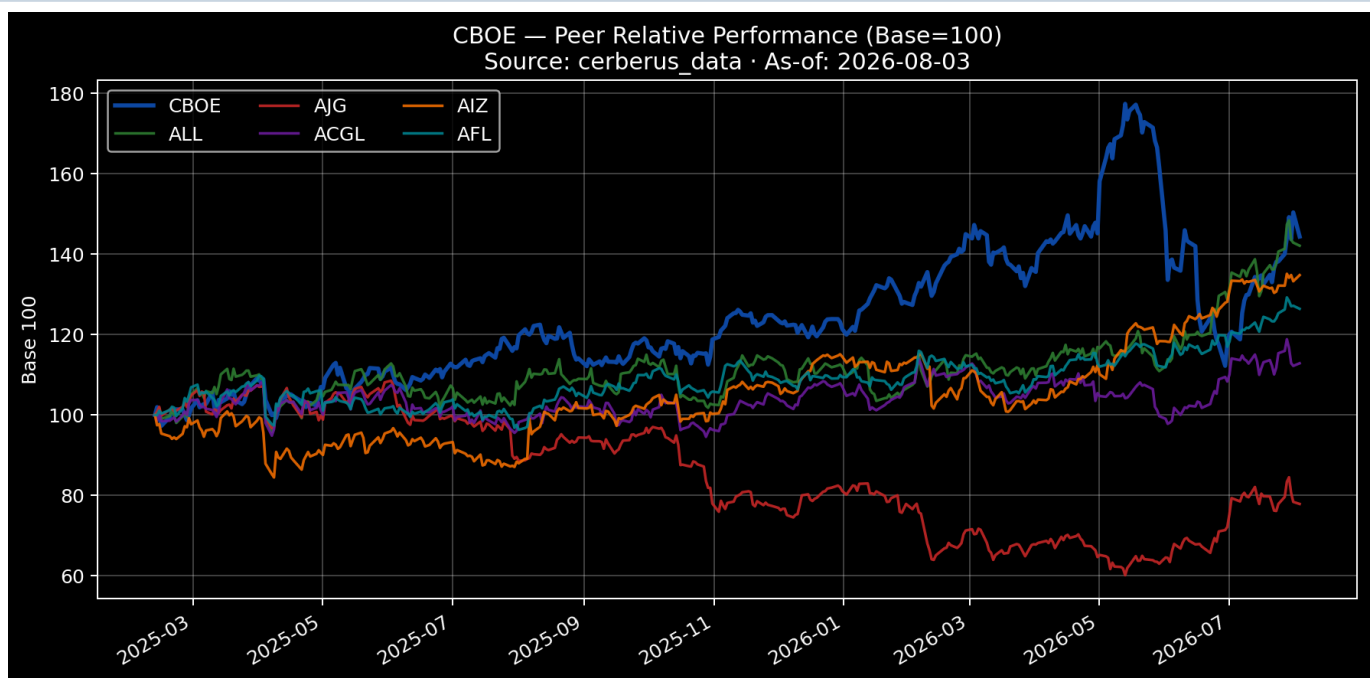


Figure 7. Peer Relative Performance (1Y, Base=100)

Read: relative performance sits near the 0th peer percentile. Why it matters: frames the allocation decision versus comparable alternatives. Source: peer snapshot / local market data · As-of: 2026-08-03

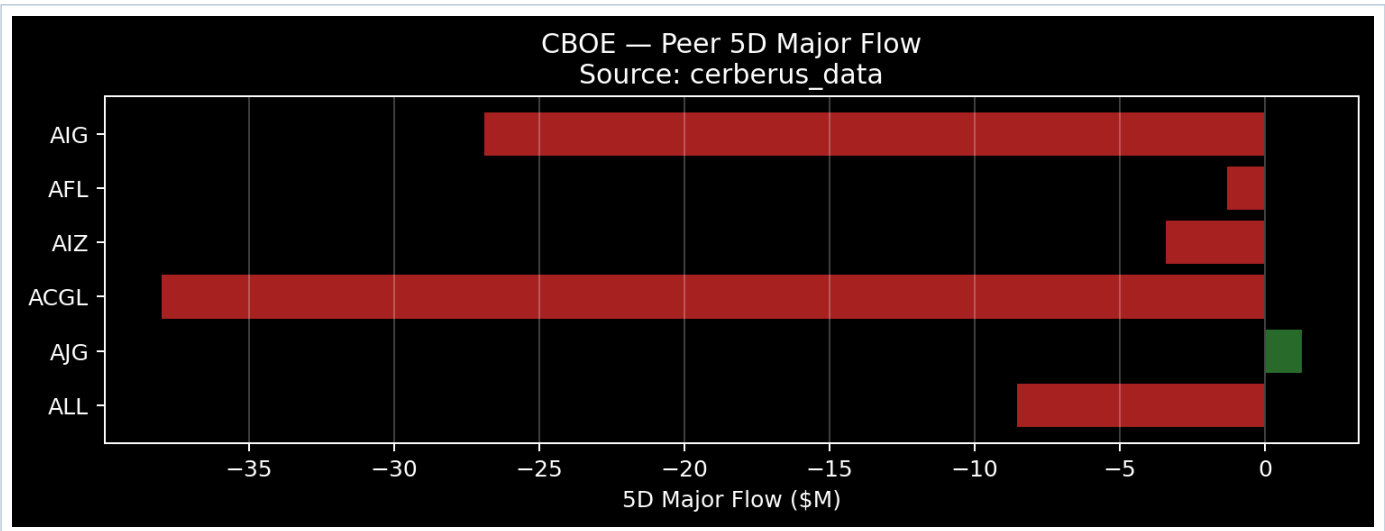


Figure 8. Peer 5D Major Flow

Why it matters: highlights where institutional flow is concentrating inside the peer set. Source: peer snapshot flows · As-of: 2026-07-31

## Flow / Positioning Dashboard

|                               |                                          |
|-------------------------------|------------------------------------------|
| AS OF<br>2026-07-31           | MAJOR NET FLOW<br>\$2.41M                |
| RETAIL NET FLOW<br>\$-4.00M   | MEDIUM NET FLOW<br>\$-3.57M              |
| SMALL NET FLOW<br>\$-427.45K  | 5D MAJOR NET FLOW<br>\$-3.32M            |
| 5D RETAIL NET FLOW<br>\$8.11M | MAJOR INFLOW RATIO<br>0.12               |
| MAJOR OUTFLOW RATIO<br>0.09   | RETAIL INFLOW RATIO<br>0.37              |
| RETAIL OUTFLOW RATIO<br>0.42  | MAJOR DAILY Z-SCORE (20D)<br>0.82        |
| SOURCE<br>cerberus_data       | FLOW TREND<br>Institutional distribution |

| Peer | 5D Major Flow |
|------|---------------|
| AJG  | \$1.26M       |
| AFL  | \$-1.29M      |
| AIZ  | \$-3.41M      |

## Flow Insight

| Lens        | Insight                                                   |
|-------------|-----------------------------------------------------------|
| Read        | flows deteriorating                                       |
| Implication | price strength needs confirmation from institutional flow |

Research only. Not investment advice. Past performance is not indicative of future results.

| Lens       | Insight                                                 |
|------------|---------------------------------------------------------|
| Watch Next | watch for price/major-flow divergence to widen or close |
| Confidence | medium / conflicting signals                            |

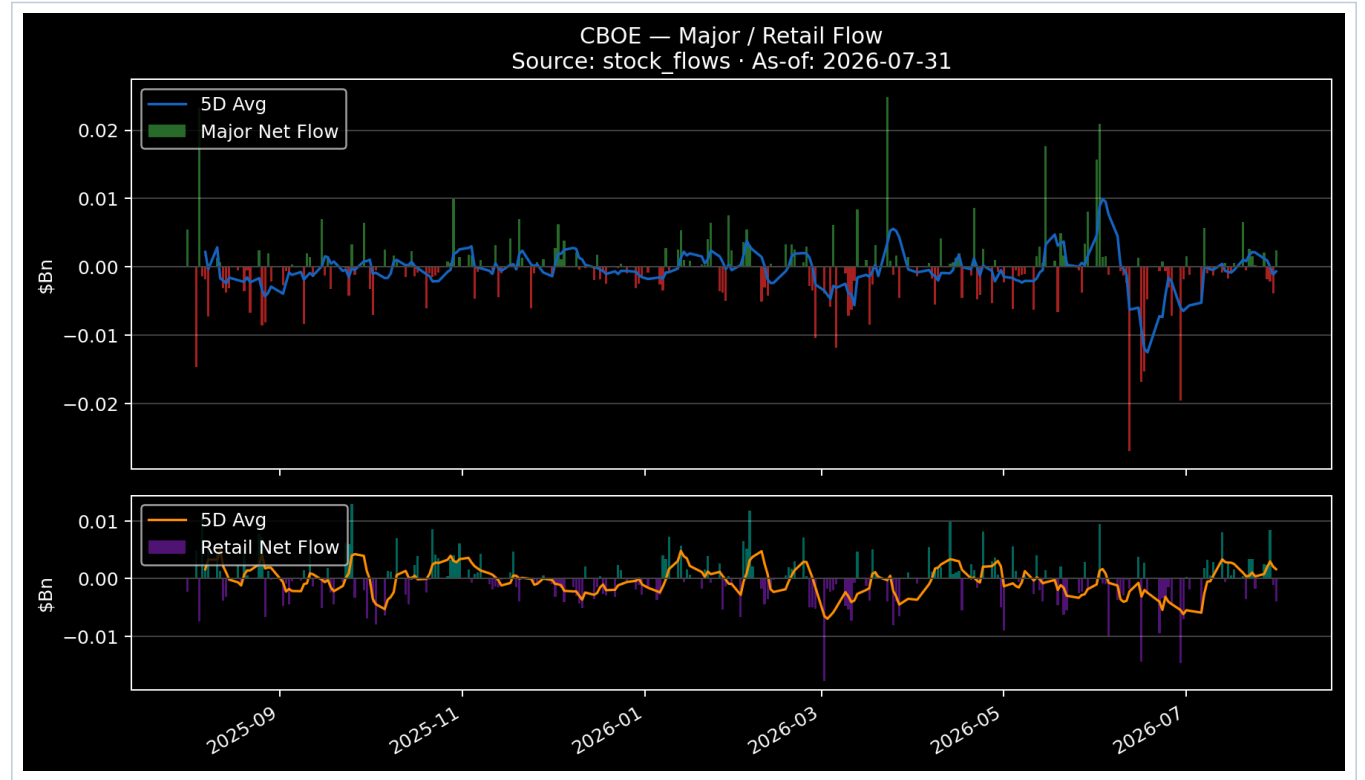


Figure 9. Major and Retail Flow Trend

Chart Read: Shows whether institutional and retail flows are confirming or fighting the tape.

Read: institutional flow deteriorating. Why it matters: shows whether positioning is confirming or fading the tape. Source: stock\_flows · As-of: 2026-07-31

AAII Retail Sentiment

|                            |                        |
|----------------------------|------------------------|
| AS OF<br>2026-07-23        | BULLISH<br>29.59%      |
| NEUTRAL<br>28.06%          | BEARISH<br>42.35%      |
| BULL-BEAR SPREAD<br>-12.76 | READ<br>retail bearish |

CBOE Volatility & Risk Premium

|                     |                                    |
|---------------------|------------------------------------|
| AS OF<br>2026-07-31 | VIX9D<br>16.38                     |
| VIX<br>7.69         | VIX3M<br>19.02                     |
| SKEW<br>141.23      | TERM SLOPE (VIX3M - VIX9D)<br>2.64 |

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BXM  
2490.46

PUT INDEX  
3557.59

| Index    | Value   |
|----------|---------|
| LONGVOL  | 5.31    |
| LOVOL    | 566.03  |
| LTV      | 8.00    |
| SHORTVOL | 1766.07 |
| SMILE    | 2526.34 |

Read: complacency

Macro / Liquidity / Positioning Dashboard

STATE  
neutral

OVERALL SCORE  
0.19

AS OF  
2026-07-31

| Component      | Score |
|----------------|-------|
| Liquidity      | 0.25  |
| Volatility     | -0.33 |
| Breadth        | 1.00  |
| Crowding       | 0.25  |
| Rates          | 0.00  |
| Macro Revision | 0.00  |

Neutral tape; prioritize selectivity and benchmark-relative confirmation.

AS OF  
2026-07-31

TAPE STATE  
Neutral

TOTAL PUT/CALL  
0.91

EQUITY PUT/CALL  
0.63

INDEX PUT/CALL  
1.01

SKEW  
141.23

VIX3M CLOSE  
19.02

SPOTVOL CLOSE  
7.69

VOL TERM PREMIUM  
11.33

forward vol above spot

AS OF  
2026-07-31

NET LIQUIDITY  
\$5.83K



|                                                     |                                                  |
|-----------------------------------------------------|--------------------------------------------------|
| <b>CREDIT SPREAD</b><br>0.80                        | <b>HY SPREAD</b><br>2.84                         |
| <b>YIELD CURVE</b><br>0.47                          | <b>FINANCIAL STRESS</b><br>-0.83                 |
| <b>NFCI</b><br>-0.55                                | <b>10Y TERM PREMIUM</b><br>0.84                  |
| <b>INTERPRETATION</b><br>net liquidity positive     |                                                  |
| <b>NAAIM AS OF</b><br>2026-07-22                    | <b>NAAIM EXPOSURE</b><br>84.02                   |
| <b>NAAIM RANGE</b><br>-25.00 / 200.00               | <b>NAAIM QUANTILES</b><br>85.00 / 95.50 / 100.00 |
| <b>NAAIM DISPERSION</b><br>43.61                    | <b>DIX/GEX AS OF</b><br>2026-07-31               |
| <b>DIX</b><br>0.46                                  | <b>GEX</b><br>\$5.74B                            |
| <b>DIX Z-SCORE (20D)</b><br>0.82                    | <b>GEX Z-SCORE (20D)</b><br>0.15                 |
| <b>READ</b><br>Crowded long; dark-pool accumulation |                                                  |

Macro Insight

| Lens        | Insight                                                               |
|-------------|-----------------------------------------------------------------------|
| Read        | macro backdrop mixed                                                  |
| Implication | macro should not be the only driver                                   |
| Watch Next  | watch regime state, liquidity stress, and sentiment crowding together |
| Confidence  | high                                                                  |

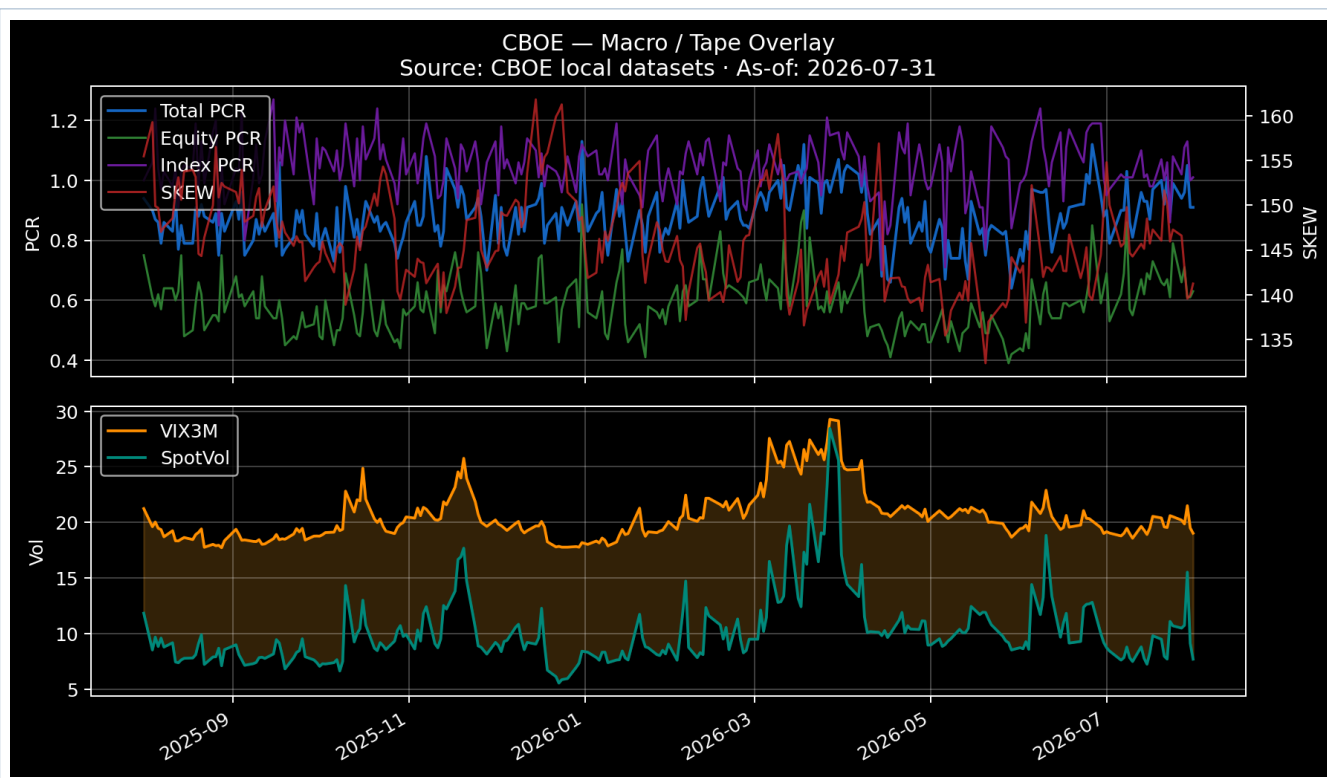


Figure 10. Macro and Tape Overlay

Why it matters: macro positioning can amplify or cap setup follow-through. Source: CBOE local datasets · As-of: 2026-07-31

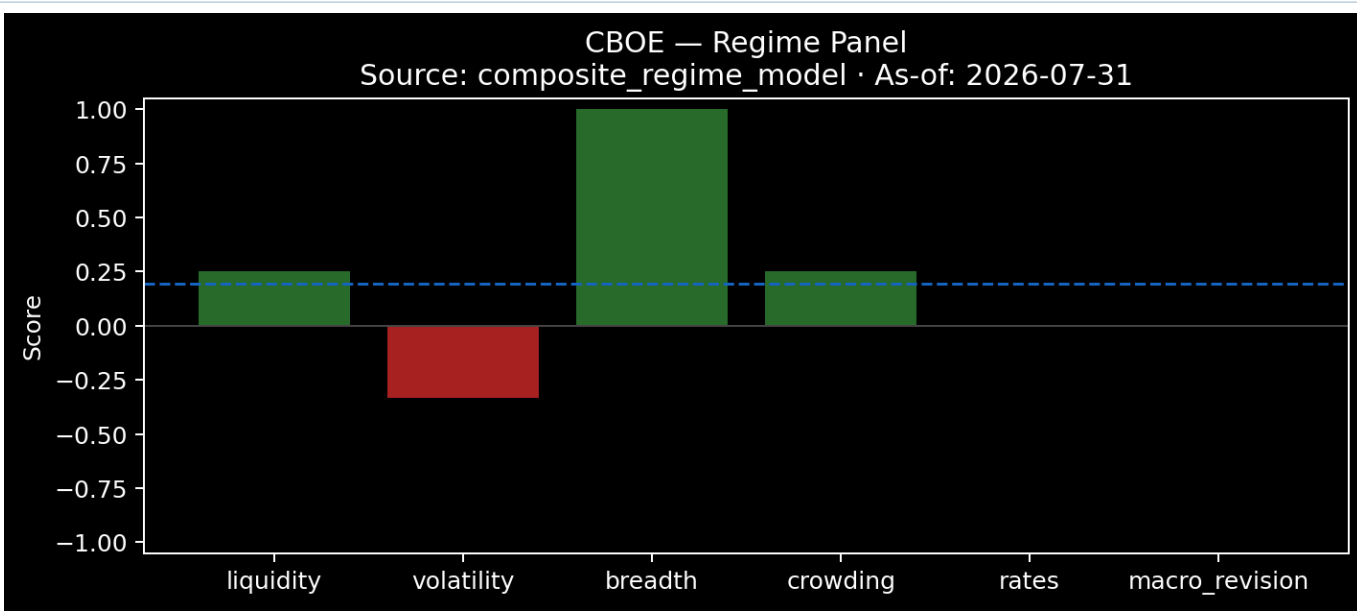


Figure 11. Market Regime Panel

Chart Read: Summarizes the macro regime drivers supporting or pressuring risk appetite.

Why it matters: regime alignment changes expected payoff and sizing discipline. Source: regime dataset · As-of: 2026-07-31

## Technical Overview

SCORE  
**52.45/100**

VERDICT  
**MONITOR**

REGIME

High-Vol Range

TREND

Uptrend

ADVICE

Hold / Neutral

CONFIDENCE

4.91%

Price vs Moving Averages

| Period           | Position  | Level    |
|------------------|-----------|----------|
| SMA 20           | Above SMA | \$282.15 |
| SMA 50           | Above SMA | \$283.06 |
| SMA 200          | Above SMA | \$279.26 |
| VWAP             | Above SMA | \$141.41 |
| Supertrend(10,3) | Above SMA | \$267.62 |

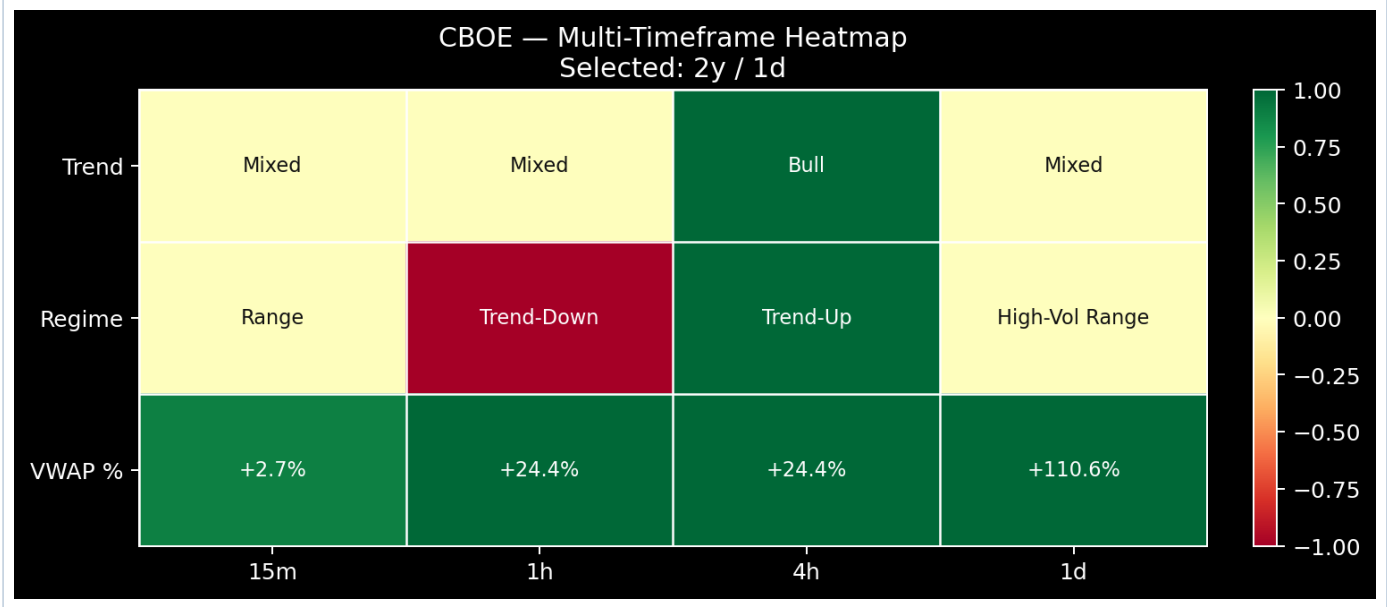


Figure 12. Multi-Timeframe Heatmap

Chart Read: Shows alignment or conflict across short and longer timeframes.

Why it matters: higher timeframe alignment improves follow-through odds. Source: technical indicators · As-of: 2026-08-03

Technical Scorecard

Setup Classification

PRIMARY SETUP

Breakout

REASON

trend 100, VCP 29, vol 57

TREND / REGIME

Uptrend / High-Vol Range

VERDICT / ADVICE

MONITOR / Hold / Neutral

STRUCTURE STAGE

Stage 2 / Markup

TREND STACK

Bull stack

VALUE POSTURE  
1/4 above

VCP / VOLATILITY STATE  
29.0 / Expansion

Execution Card

TRIGGER  
>311.06 or reclaim 294.88

INVALIDATION  
<272.25 / EMA21 285.37

TARGET  
348.99

REWARD / RISK  
2.0:1

TIMEFRAME BIAS  
Bull

Watch Conditions

| Trigger             | Level / Status                |
|---------------------|-------------------------------|
| Donchian Breakout   | >311.06                       |
| VWAP Posture        | +110.6%                       |
| EMA9 Reclaim/Hold   | 294.88                        |
| EMA21 Reclaim/Hold  | 285.37                        |
| Target              | 348.99                        |
| Volume Confirmation | 57.5                          |
| Compression Flags   | NR7 No / Inside Yes / TTM Off |

Invalidation Logic

| Check            | Threshold |
|------------------|-----------|
| ATR Stop         | <272.25   |
| Donchian Lower   | <244.35   |
| Supertrend       | <267.62   |
| EMA21            | <285.37   |
| Current Drawdown | -18.8%    |
| Risk / Reward    | 2.0:1     |

Timeframe Agreement

| Interval  | Trend   | Regime         | Price vs VWAP |
|-----------|---------|----------------|---------------|
| 15m       | Mixed   | Range          | +2.7%         |
| 1h        | Mixed   | Trend-Down     | +24.4%        |
| 4h        | Bull    | Trend-Up       | +24.4%        |
| 1d        | Mixed   | High-Vol Range | +110.6%       |
| Consensus | 100.00% | Bull           | —             |

Research only. Not investment advice. Past performance is not indicative of future results.

Regime / Driver Breakdown

| Driver              | Value |
|---------------------|-------|
| Final Score         | 52.5  |
| Confidence          | 4.9   |
| Momentum            | 69.2  |
| Trend               | 100.0 |
| Risk Adjusted       | 42.4  |
| Vol Stability       | 57.1  |
| Drawdown            | 0.0   |
| Volume Confirmation | 57.5  |
| ATR Percentile      | 92.5  |
| Vol Clustering      | 0.358 |
| Gap Fill Prob       | 90.4% |

Technical Insight

| Lens        | Insight                                                  |
|-------------|----------------------------------------------------------|
| Read        | technicals aligned                                       |
| Implication | current structure supports acting on strength            |
| Watch Next  | breakout trigger, MTF bias Bull, and volume confirmation |
| Confidence  | high                                                     |

Market Structure Deep Dive

Support / Resistance Clusters

| Side       | Cluster                              |
|------------|--------------------------------------|
| Resistance | 302.94 (+1.7%) x1 [Pivot]            |
| Support    | 294.88 (-1.0%) x3 [EMA9, Fib S1, S1] |

Distance Matrix

| Level  | Value  | Distance |
|--------|--------|----------|
| VWAP   | 141.41 | +110.6%  |
| EMA9   | 294.88 | +1.0%    |
| EMA21  | 285.37 | +4.4%    |
| SMA50  | 283.06 | +5.2%    |
| SMA200 | 279.26 | +6.7%    |
| Pivot  | 302.94 | -1.7%    |

Research only. Not investment advice. Past performance is not indicative of future results.

| Level          | Value  | Distance |
|----------------|--------|----------|
| R1             | 318.35 | -6.4%    |
| S1             | 294.83 | +1.0%    |
| Donchian Upper | 311.06 | -4.3%    |
| Donchian Lower | 244.35 | +21.9%   |
| Stop           | 272.25 | +9.4%    |
| Target         | 348.99 | -14.7%   |

Candle / Bar Anatomy

|                                                    |                                                |
|----------------------------------------------------|------------------------------------------------|
| O / H / L / C<br>302.53 / 310.75 / 292.47 / 297.83 | BODY / UPPER WICK / LOWER WICK<br>26 / 45 / 29 |
| RANGE / ATR<br>18.28 / 1.43x                       | CLOSING RANGE<br>29.3%                         |
| STREAK<br>-1                                       |                                                |

Historical Behavior Near Key MAs

| MA     | Behavior  |
|--------|-----------|
| DMA20  | avg +1.0% |
| DMA50  | avg +0.9% |
| DMA200 | avg +0.2% |

Compression / Expansion

| Flag                            | Value        |
|---------------------------------|--------------|
| VCP Score                       | 29.0         |
| Volatility State                | Expansion    |
| TTM Squeeze                     | Off          |
| NR7 / Inside Bar                | No / Yes     |
| ATR Percentile / Vol Clustering | 92.5 / 0.358 |

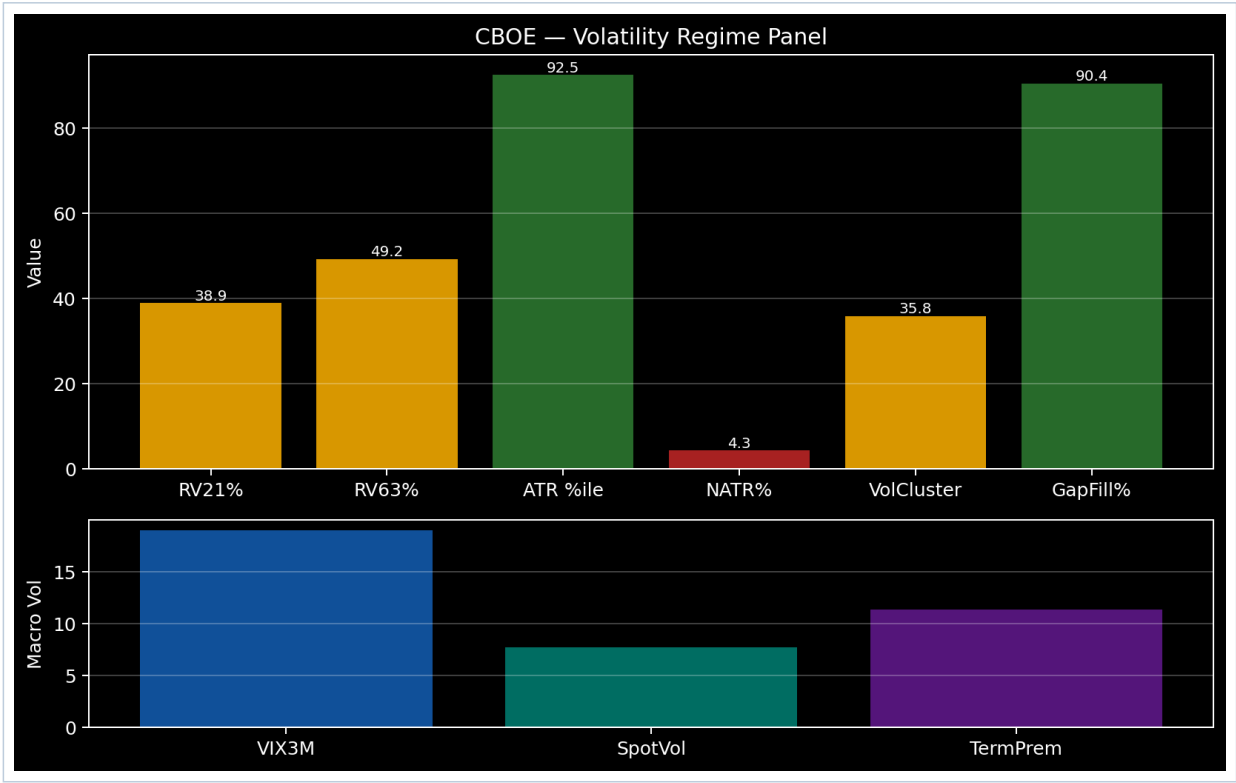


Figure 13. Volatility Regime Panel

Chart Read: Shows whether realized and implied volatility conditions favor expansion or compression.

Read: ATR is running near 4.3% of spot, so wider stops may be necessary. Why it matters: vol state affects breakout odds and stop placement. Source: technical indicators · As-of: 2026-08-03

Technical Indicators

| Indicator   | Value   | Indicator   | Value       |
|-------------|---------|-------------|-------------|
| RSI(14)     | 57.68   | Stoch K/D   | 78.71/85.09 |
| MACD        | 7.46    | MACD Signal | 3.72        |
| MACD Hist   | +3.74   | Williams %R | -32.39      |
| ADX(14)     | 20.47   | CCI(20)     | 118.05      |
| ATR(14)     | 12.79   | NATR(14)%   | 4.29%       |
| BB Upper    | 310.12  | BB Lower    | 254.19      |
| BB Position | 78.03%  | Z-Score     | 1.12        |
| ROC(21)%    | +19.62% | ROC(63)%    | -8.91%      |
| SAR         | 287.54  | ER(10)      | 0.26        |
| OBV Ratio   | 0.06    | 52W Pos%    | 49.06%      |

Pattern Signals

| Signal | Status |
|--------|--------|
| NR7    | —      |

| Signal         | Status |
|----------------|--------|
| Inside Bar     | Yes    |
| Bull Pin Bar   | —      |
| Bear Pin Bar   | —      |
| Closing Range% | 29.32% |
| Streak (bars)  | -1     |

Market Structure

|                                             |                           |
|---------------------------------------------|---------------------------|
| <b>WEINSTEIN STAGE</b><br>Stage 2 (Uptrend) | <b>VCP SCORE</b><br>28.96 |
|---------------------------------------------|---------------------------|

Risk Levels (ATR-based)

| Level       | Price    | Note               |
|-------------|----------|--------------------|
| Stop Loss   | \$272.25 | 2x ATR below entry |
| Target      | \$348.99 | 4x ATR above entry |
| Risk/Reward | 2.00:1   | —                  |

Risk & Quantitative Metrics

| Metric             | 21D   | 63D   |
|--------------------|-------|-------|
| Realized Vol (Ann) | 0.39% | 0.49% |
| Sharpe Ratio       | 5.74  | -0.51 |
| Sortino Ratio      | 11.99 | -0.73 |
| Calmar             | 55.88 | -0.68 |

|                                    |                                     |
|------------------------------------|-------------------------------------|
| <b>CURRENT DRAWDOWN</b><br>-18.78% | <b>MAX DRAWDOWN (2Y)</b><br>-44.70% |
| <b>CVAR 95%</b><br>-8.02%          | <b>KELLY FRACTION</b><br>-0.05      |
| <b>HURST EXPONENT</b><br>1.00      | <b>SKEWNESS</b><br>-0.31            |
| <b>KURTOSIS</b><br>-0.11           | <b>VOL CLUSTERING</b><br>0.36       |
| <b>GAP FILL PROB%</b><br>90.44%    | <b>MONTHLY WIN RATE%</b><br>70.00%  |
| <b>MTF CONSENSUS%</b><br>100.00%   |                                     |

Score Breakdown



| Component      | Score  |
|----------------|--------|
| Momentum       | 69.22  |
| Trend          | 100.00 |
| Mean Reversion | 29.82  |
| Bollinger      | 21.97  |
| VWAP           | 0.00   |
| Trend Strength | 40.94  |
| Vol Stability  | 57.06  |
| Risk-Adjusted  | 42.40  |
| Drawdown       | 0.00   |
| Volume         | 57.49  |

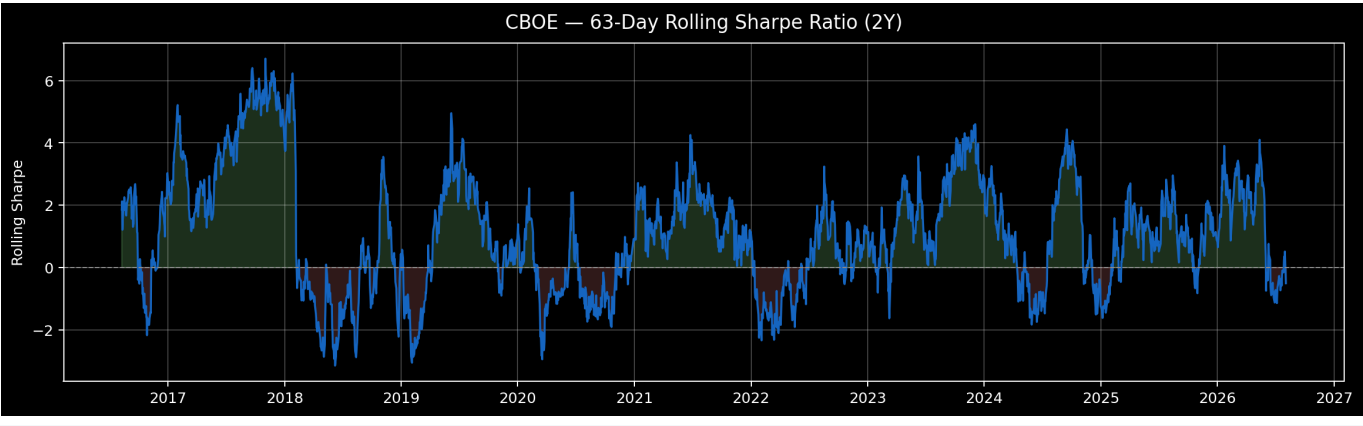


Figure 14. 63-Day Rolling Sharpe Ratio (2Y)

Chart Read: Shows whether returns have recently been efficient or unstable on a risk-adjusted basis.

Why it matters: shows whether recent returns have been efficient or unstable. Source: yfinance daily bars · As-of: 2026-08-03

**NOTE**  
Primary sources: SEC companyfacts, local actions, yfinance fallback. Derived fields are labeled.

Issuer Profile

|                                 |                                                            |
|---------------------------------|------------------------------------------------------------|
| <b>WEBSITE</b><br>cboe.com      | <b>SEC PROFILE</b><br>CIK                                  |
| <b>CIK</b><br>0001374310        | <b>LISTING</b><br>Cboe US / equity                         |
| <b>COUNTRY</b><br>United States | <b>DOMICILE / LISTING</b><br>United States / United States |
| <b>EMPLOYEES</b><br>1.66K       | <b>IPO DATE</b><br>2010-06-15                              |

ACTIVE LISTING  
True

ETF / ADR / REIT  
False / False / False

STATUS REASON  
active

Identifiers

FIGI  
BBG000QH56C1

SHARE CLASS FIGI  
BBG001T71BP7

OPENFIGI MARKET SECTOR  
Equity

COMPOSITE FIGI  
BBG000QH56C1

OPENFIGI TYPE  
Common Stock / Common Stock

OPENFIGI EXCHANGE / MIC  
US

Valuation

TRAILING P/E  
22.66

PEG RATIO  
2.93

EV/EBITDA  
15.71

MARKET CAP  
\$30.39B

FORWARD P/E  
19.06

P/S RATIO  
6.00

PRICE/BOOK  
5.41

ENTERPRISE VALUE  
\$30.33B

Profitability & Margins

GROSS MARGIN  
54.08%

NET MARGIN  
26.73%

ROA  
11.58%

GROSS PROFIT  
\$2.74B

OP MARGIN  
34.63%

ROE  
26.30%

EBITDA  
\$1.93B

Income Statement (TTM)

REVENUE  
\$5.06B

NET INCOME  
\$1.35B

OPERATING INCOME  
\$505.60M

EBITDA  
\$1.93B

Most Recent Quarter

|                           |                           |
|---------------------------|---------------------------|
| Q REVENUE<br>\$1.27B      | Q OP INCOME<br>\$505.60M  |
| Q NET INCOME<br>\$385.70M | Q OP CASH FLOW<br>\$1.96B |

Point-in-Time Trend Snapshot

| Metric              | 2026-06-30 | 2026-03-31 | 2025-12-31 | 2025-09-30 |
|---------------------|------------|------------|------------|------------|
| Revenue             | —          | \$1.27B    | \$1.20B    | \$1.14B    |
| Net Income          | —          | \$385.70M  | \$313.50M  | \$300.80M  |
| Operating Income    | —          | \$505.60M  | \$428.80M  | \$375.00M  |
| Operating Cash Flow | \$1.96B    | \$352.10M  | \$153.90M  | \$333.70M  |

Fundamental Score Summary

|                                    |                                   |
|------------------------------------|-----------------------------------|
| GROWTH SCORE<br>50.00              | QUALITY SCORE<br>72.50            |
| RESILIENCE SCORE<br>63.78          | CAPITAL ALLOCATION SCORE<br>65.00 |
| OVERALL FUNDAMENTAL SCORE<br>62.82 | OVERALL POSTURE<br>Balanced       |

Margin Trend / Growth Regime

|                                            |                                                |
|--------------------------------------------|------------------------------------------------|
| GROWTH REGIME<br>Stable compounder         | REVENUE TREND<br>insufficient history          |
| GROSS MARGIN TREND<br>insufficient history | OPERATING MARGIN TREND<br>insufficient history |
| NET MARGIN TREND<br>insufficient history   |                                                |

Balance Sheet & Resilience

|                                      |                               |
|--------------------------------------|-------------------------------|
| TOTAL ASSETS<br>\$11.07B             | TOTAL LIABILITIES<br>\$5.70B  |
| STOCKHOLDERS EQUITY<br>\$5.37B       | CASH & EQUIVALENTS<br>\$2.22B |
| TOTAL DEBT<br>\$1.58B                | NET DEBT<br>\$-636.20M        |
| STRESS READ<br>Net cash              | NET DEBT / EBITDA<br>-0.33    |
| CURRENT / QUICK RATIO<br>1.56 / 0.79 | DEBT / EQUITY<br>28.09        |

Cash Flow & Conversion

|                                                             |                                                              |
|-------------------------------------------------------------|--------------------------------------------------------------|
| <div>OPERATING CASH FLOW</div> <div>\$1.96B</div>           | <div>CAPITAL EXPENDITURES (DERIVED)</div> <div>\$1.28B</div> |
| <div>FREE CASH FLOW</div> <div>\$678.25M</div>              | <div>OCF / NET INCOME</div> <div>1.46</div>                  |
| <div>FCF / NET INCOME</div> <div>0.50</div>                 | <div>CAPEX INTENSITY (DERIVED)</div> <div>25.38%</div>       |
| <div>EARNINGS QUALITY</div> <div>Cash-backed earnings</div> |                                                              |

Ownership & Short Interest

|                                             |                                              |
|---------------------------------------------|----------------------------------------------|
| <div>INSIDER HOLDING</div> <div>0.25%</div> | <div>INST. HOLDING</div> <div>92.22%</div>   |
| <div>SHORT % FLOAT</div> <div>2.98%</div>   | <div>SHORT RATIO (DTC)</div> <div>2.87</div> |
| <div>SHARES SHORT</div> <div>2.55M</div>    |                                              |

Growth & Yield

|                                             |                                              |
|---------------------------------------------|----------------------------------------------|
| <div>REVENUE GROWTH</div> <div>22.90%</div> | <div>EARNINGS GROWTH</div> <div>50.20%</div> |
| <div>FCF YIELD</div> <div>2.23%</div>       | <div>PAYOUT RATIO</div> <div>22.45%</div>    |
| <div>EPS FWD</div> <div>15.27</div>         |                                              |

Capital Returns & Share Count

|                                                        |                                                                        |
|--------------------------------------------------------|------------------------------------------------------------------------|
| <div>DIVIDEND YIELD</div> <div>93.00%</div>            | <div>PAYOUT RATIO</div> <div>22.45%</div>                              |
| <div>DIVIDEND CADENCE</div> <div>quarterly</div>       | <div>LATEST DIVIDEND</div> <div>2026-05-25</div>                       |
| <div>SHARES OUTSTANDING</div> <div>104.43M</div>       | <div>SHARE COUNT TREND</div> <div>insufficient history</div>           |
| <div>SHAREHOLDER YIELD (EST.)</div> <div>93.00%</div>  | <div>YIELD DECOMPOSITION</div> <div>dividend 93.00% + buyback -%</div> |
| <div>BUYBACK / DILUTION REGIME</div> <div>Stable</div> |                                                                        |

Valuation Context

|                                                 |                                                |
|-------------------------------------------------|------------------------------------------------|
| <div>TRAILING P/E CONTEXT</div> <div>fair</div> | <div>FORWARD P/E CONTEXT</div> <div>fair</div> |
|-------------------------------------------------|------------------------------------------------|

Research only. Not investment advice. Past performance is not indicative of future results.

EV / EBITDA CONTEXT

fair

FCF YIELD CONTEXT

thin

OVERALL VALUATION POSTURE

Mixed

COVERAGE NOTE

peer/sector valuation history unavailable in current snapshot

Sector-Relative Fundamentals

NET MARGIN PERCENTILE

not meaningful

OPERATING MARGIN PERCENTILE

not meaningful

FUNDAMENTAL MOMENTUM SCORE

62.82

RELATIVE READ

insufficient peer sample

Fundamentals Insight

| Lens        | Insight                                                              |
|-------------|----------------------------------------------------------------------|
| Read        | balanced                                                             |
| Implication | fundamentals are serviceable but not dominant                        |
| Watch Next  | watch growth regime, cash conversion, and share-count drift together |
| Confidence  | medium                                                               |

Valuation Sanity Flags

TRAILING PE

22.66

FORWARD PE

19.06

EV / EBITDA

15.71

PRICE / SALES

6.00

DIVIDEND YIELD

93.00%

FLAG COUNT

0

Fundamental Quality Panel

QUALITY SCORE

87.56

VERDICT

Strong

GROWTH

revenue 22.90%, earnings 50.20%

MARGINS

gross 54.08%, operating 34.63%, net 26.73%

FCF YIELD

2.23%

LIQUIDITY

current 1.56, quick 0.79

DEBT / EQUITY

28.09

Fundamental Quality Insight

| Lens | Insight                    |
|------|----------------------------|
| Read | fundamental quality strong |

Research only. Not investment advice. Past performance is not indicative of future results.

| Lens        | Insight                                                          |
|-------------|------------------------------------------------------------------|
| Implication | fundamentals support paying up for the name                      |
| Watch Next  | watch growth, FCF yield, and leverage move in the same direction |
| Confidence  | high                                                             |

Curated Fundamentals

|                                                  |                                                                                                                                                       |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SYMBOL</b><br>CBOE                            | <b>FISCAL PERIOD END</b><br>2026-03-31                                                                                                                |
| <b>FISCAL YEAR</b><br>2026                       | <b>FISCAL QUARTER</b><br>1                                                                                                                            |
| <b>PERIOD TYPE</b><br>quarterly                  | <b>RUN ID</b><br>52d6acc474838516                                                                                                                     |
| <b>COMPANY NAME</b><br>Cboe Global Markets, Inc. | <b>DESCRIPTION</b><br>Cboe Global Markets, Inc., through its subsidiaries, operates as a derivatives and securities exchange network that delivers... |
| <b>SECTOR</b><br>Financial Services              | <b>INDUSTRY</b><br>Financial Data & Stock Exchanges                                                                                                   |
| <b>COUNTRY</b><br>United States                  | <b>WEBSITE</b><br><a href="https://www.cboe.com">https://www.cboe.com</a>                                                                             |
| <b>FULL TIME EMPLOYEES</b><br>1661               | <b>CURRENCY</b><br>USD                                                                                                                                |
| <b>EXCHANGE</b><br>BTS                           | <b>SHARES OUTSTANDING</b><br>104653784                                                                                                                |
| <b>FLOAT SHARES</b><br>104228890                 | <b>HELD PERCENT INSTITUTIONS</b><br>0.92                                                                                                              |
| <b>HELD PERCENT INSIDERS</b><br>0.00             | <b>PROFIT MARGINS</b><br>0.26                                                                                                                         |
| <b>OPERATING MARGINS</b><br>0.40                 | <b>GROSS MARGINS</b><br>0.54                                                                                                                          |
| <b>EBITDA MARGINS</b><br>0.37                    | <b>RETURN ON ASSETS</b><br>0.11                                                                                                                       |
| <b>RETURN ON EQUITY</b><br>0.25                  | <b>REVENUE GROWTH</b><br>0.07                                                                                                                         |
| <b>EARNINGS GROWTH</b><br>0.54                   | <b>EARNINGS QUARTERLY GROWTH</b><br>0.54                                                                                                              |
| <b>TRAILING EPS</b><br>11.70                     | <b>FORWARD EPS</b><br>14.72                                                                                                                           |
| <b>DIVIDEND RATE</b><br>2.88                     | <b>DIVIDEND YIELD</b><br>1.02                                                                                                                         |

|                                                                                                                                                               |                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>PAYOUT RATIO</b><br>0.24                                                                                                                                   | <b>HAS ANALYST PRICE TARGETS</b><br>True                                                |
| <b>RECOMMENDATIONS ROWS</b><br>4                                                                                                                              | <b>UPGRADES DOWNGRADES ROWS</b><br>328                                                  |
| <b>CALENDAR EVENT COUNT</b><br>0                                                                                                                              | <b>HISTORY METADATA HAS DATA</b><br>True                                                |
| <b>INCOME TOTAL REVENUE</b><br>1272800000.00                                                                                                                  | <b>INCOME GROSS PROFIT</b><br>601000000.00                                              |
| <b>INCOME OPERATING INCOME</b><br>505600000.00                                                                                                                | <b>INCOME NET INCOME</b><br>385700000.00                                                |
| <b>BALANCE TOTAL ASSETS</b><br>11070600000.00                                                                                                                 | <b>BALANCE TOTAL LIABILITIES</b><br>5697100000.00                                       |
| <b>BALANCE TOTAL EQUITY</b><br>5373500000.00                                                                                                                  | <b>CASHFLOW OPERATING CASHFLOW</b><br>1960000000.00                                     |
| <b>CASHFLOW FREE CASHFLOW</b><br>1940800000.00                                                                                                                | <b>SOURCE USED FINANCIALS</b><br>yahoo                                                  |
| <b>ENDPOINTS COLLECTED</b><br>["analyst_price_targets", "balance_sheet", "calendar", "cashflow", "fast_info", "history_metadata", "income_stmt", "info", ...] | <b>PAYLOAD HASH</b><br>07f7fd631a697fa5eb94a11b57d30a94ddc0083dcbac9422a7212b3994a6fe3e |
| <b>CURATED LOGIC VERSION</b><br>2                                                                                                                             | <b>COLLECTED AT</b><br>2026-06-07 18:21 UTC                                             |
| <b>INCOME TOTAL REVENUE LATEST</b><br>1272800000.00                                                                                                           | <b>INCOME GROSS PROFIT LATEST</b><br>601000000.00                                       |
| <b>INCOME OPERATING INCOME LATEST</b><br>505600000.00                                                                                                         | <b>INCOME NET INCOME LATEST</b><br>385700000.00                                         |
| <b>BALANCE TOTAL ASSETS LATEST</b><br>11070600000.00                                                                                                          | <b>BALANCE TOTAL LIABILITIES LATEST</b><br>5697100000.00                                |
| <b>BALANCE TOTAL EQUITY LATEST</b><br>5373500000.00                                                                                                           | <b>CASHFLOW OPERATING CASHFLOW LATEST</b><br>1960000000.00                              |
| <b>CASHFLOW FREE CASHFLOW LATEST</b><br>1940800000.00                                                                                                         | <b>FINANCIALS AS OF DATE</b><br>2026-03-31                                              |

Corporate Actions

|                                                                                                            |
|------------------------------------------------------------------------------------------------------------|
| <b>NOTE</b><br>Charts and relative returns use adjusted close when available; raw OHLC remains unadjusted. |
|------------------------------------------------------------------------------------------------------------|

Recent Dividends

| Date       | Amount | Source   |
|------------|--------|----------|
| 2026-05-25 | 0.72   | yfinance |

Research only. Not investment advice. Past performance is not indicative of future results.

| Date       | Amount | Source   |
|------------|--------|----------|
| 2026-02-23 | 0.72   | yfinance |

## Earnings & Revenue

### Next Earnings

| Next Date  | EPS Estim... | Revenue Estimate |
|------------|--------------|------------------|
| 2026-10-30 | 3.47         | \$707.51M        |

### Estimate Range

| Metric           | Low       | High      |
|------------------|-----------|-----------|
| EPS Estimate     | 3.06      | 3.56      |
| Revenue Estimate | \$659.10M | \$723.00M |

### EPS Surprise History (Last 8 Quarters)

| Date       | EPS Estim... | Reported E... | Surprise    |
|------------|--------------|---------------|-------------|
| 2026-10-30 | 3.47         | —             | —           |
| 2026-07-31 | 3.49         | 3.56          | OK 2.01%    |
| 2026-05-01 | 3.38         | 3.70          | OK 9.54%    |
| 2026-02-06 | 2.95         | 3.06          | OK 3.76%    |
| 2025-10-31 | 2.53         | 2.67          | OK 5.58%    |
| 2025-08-01 | 2.43         | 2.46          | OK 1.11%    |
| 2025-05-02 | 2.36         | 2.50          | OK 5.76%    |
| 2025-02-07 | 2.12         | 2.10          | DOWN -0.88% |

### Annual Revenue & Net Income

| Year | Revenue | Net Income | Net Margin |
|------|---------|------------|------------|
| 2025 | \$4.71B | \$1.10B    | 23.33%     |
| 2024 | \$4.09B | \$764.90M  | 18.68%     |
| 2023 | \$3.77B | \$761.40M  | 20.18%     |
| 2022 | \$3.96B | \$235.00M  | 5.94%      |
| 2021 | —       | —          | —          |

### Share Buybacks (Annual)

| Year | Repurchases |
|------|-------------|
| 2025 | \$66.70M    |



| Year | Repurchases |
|------|-------------|
| 2024 | \$204.80M   |
| 2023 | \$83.90M    |
| 2022 | \$100.90M   |

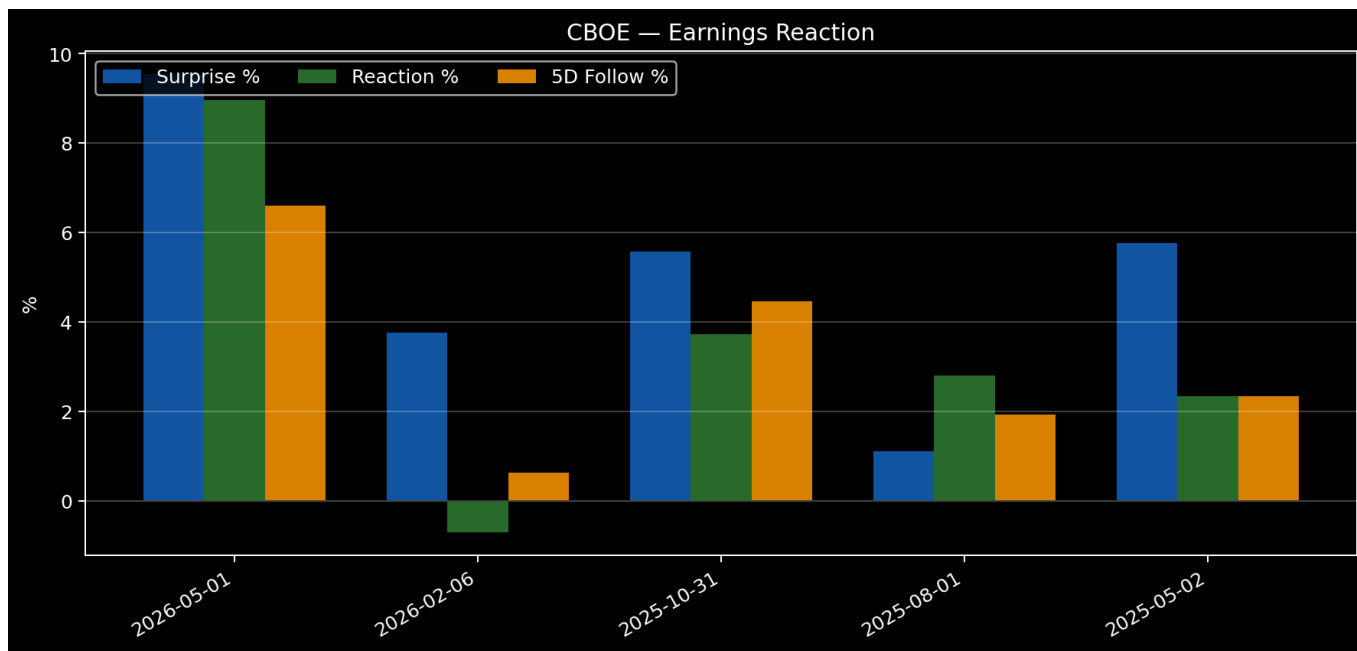


Figure 15. Earnings Reaction

Chart Read: Shows how the stock has typically moved around earnings events.

Why it matters: event reaction history sets expectations into the next print. Source: earnings history · As-of: 2025-02-07

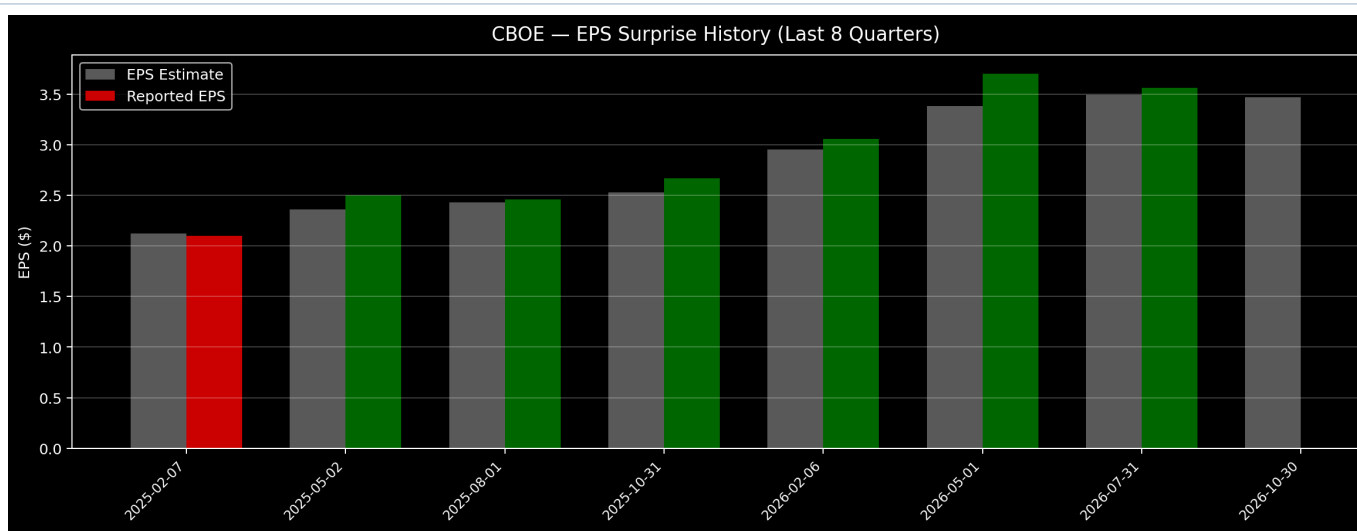


Figure 16. EPS Surprise History

Chart Read: Shows whether earnings have tended to beat or miss expectations.

Why it matters: surprise consistency helps separate narrative from execution. Source: earnings history · As-of: 2025-02-07

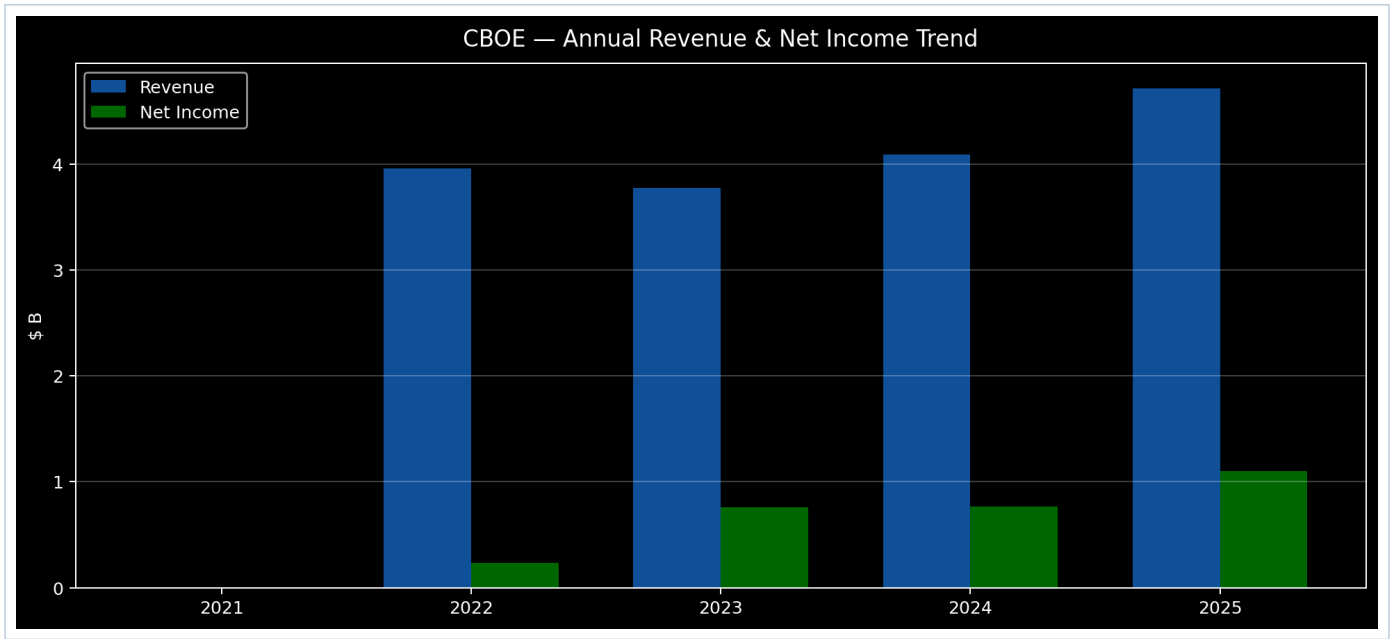


Figure 17. Annual Revenue & Net Income Trend

Chart Read: Shows the slope and consistency of top-line growth and profitability.

Why it matters: slope and consistency matter more than one quarter in isolation. Source: annual revenue history · As-of: 1970-01-01

Options Snapshot

|                              |                          |
|------------------------------|--------------------------|
| NEAREST EXPIRY<br>2026-08-07 | ATM IV (CALLS)<br>49.31% |
| ATM IV (PUTS)<br>49.33%      | PUT/CALL RATIO<br>0.26   |
| TOTAL CALL OI<br>1.30K       | TOTAL PUT OI<br>1.38K    |

Options Expected Move

|                                   |                                   |
|-----------------------------------|-----------------------------------|
| FRONT EXPIRY<br>2026-08-07        | DTE<br>3                          |
| ATM IV USED<br>49.31%             | EXPECTED MOVE<br>+/-16.02 (5.38%) |
| IMPLIED RANGE<br>281.81 to 313.85 |                                   |

Analyst Consensus

|                           |                                     |
|---------------------------|-------------------------------------|
| RECOMMENDATION<br>HOLD    | ANALYST COUNT<br>14                 |
| TARGET (MEAN)<br>\$316.00 | TARGET RANGE<br>\$258.00 - \$380.00 |

| Rating      | Count | Share        |
|-------------|-------|--------------|
| Strong Buy  | 3     | ## 17.65%    |
| Buy         | 1     | # 5.88%      |
| Hold        | 10    | ##### 58.82% |
| Sell        | 2     | # 11.76%     |
| Strong Sell | 1     | # 5.88%      |

## #### Recent Upgrades / Downgrades

| Date       | Firm                    | Action    | Grade          |
|------------|-------------------------|-----------|----------------|
| 2026-05-04 | UBS                     | Reiterate | Neutral        |
| 2026-05-04 | TD Cowen                | Reiterate | Hold           |
| 2026-05-04 | Barclays                | Reiterate | Overweight     |
| 2026-05-04 | Keefe, Bruyette & Woods | Reiterate | Market Perform |
| 2026-04-15 | Piper Sandler           | Reiterate | Overweight     |
| 2026-04-10 | Morgan Stanley          | Reiterate | Underweight    |
| 2026-04-08 | Barclays                | Reiterate | Overweight     |
| 2026-02-09 | Barclays                | Reiterate | Overweight     |

## Analyst Insight

| Lens        | Insight                                                          |
|-------------|------------------------------------------------------------------|
| Read        | street view mixed                                                |
| Implication | consensus is not a strong differentiator                         |
| Watch Next  | watch target dispersion and rating breadth for revision pressure |
| Confidence  | medium / conflicting signals                                     |

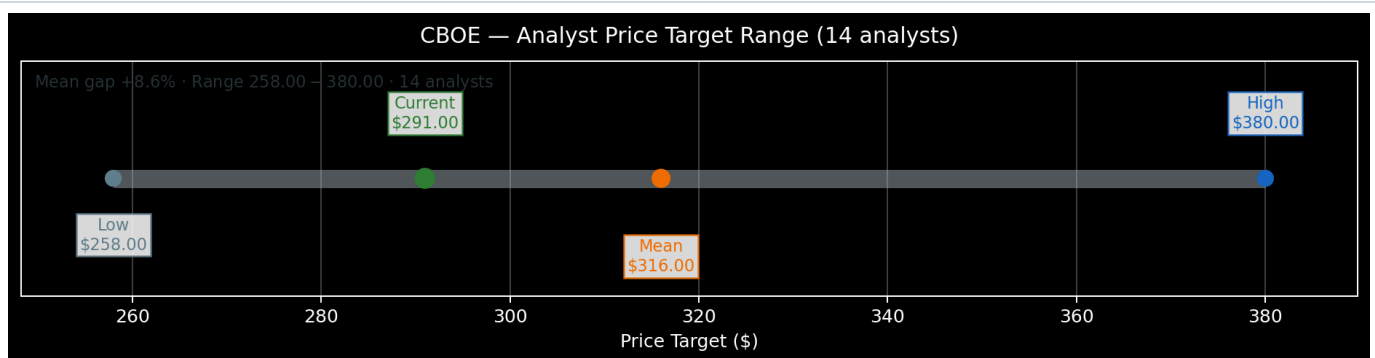


Figure 18. Analyst Price Target Range

Chart Read: Shows how optimistic or conservative the street target range currently is.

Read: consensus implies +6.1% with 38.6% target spread. Why it matters: shows how much of the upside case is already in consensus. Source: analyst consensus · As-of: 2026-08-03

Ownership / Crowding Context

|                                                 |                                                                                              |
|-------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>CROWDING READ</b><br>Distribution risk       | <b>TOP INSTITUTIONAL HOLDER</b><br>Blackrock Inc.                                            |
| <b>TOP INSTITUTIONAL VALUE</b><br>\$2.55B       | <b>TOP-5 INSTITUTIONAL CONCENTRATION</b><br>26.02%                                           |
| <b>INSTITUTIONAL HOLDER COUNT</b><br>5          | <b>TOP MUTUAL FUND HOLDER</b><br>VANGUARD INDEX FUNDS-Vanguard Total Stock Market Index Fund |
| <b>TOP-5 MUTUAL FUND CONCENTRATION</b><br>8.73% | <b>MUTUAL FUND % OUT (TOP-5)</b><br>9.79%                                                    |
| <b>RECENT INSIDER BUYS / SELLS</b><br>0 / 2     | <b>RECENT INSIDER NET VALUE</b><br>\$-510.56K                                                |

Ownership Insight

| Lens        | Insight                                                  |
|-------------|----------------------------------------------------------|
| Read        | distribution risk                                        |
| Implication | holder base can amplify downside if conviction breaks    |
| Watch Next  | watch insider net activity against concentration changes |
| Confidence  | high                                                     |

Ownership Change Over Time

| Metric                       | Previous | Current | Delta  |
|------------------------------|----------|---------|--------|
| insidersPercentHeld          | 0.25%    | 0.25%   | +0.00% |
| institutionsPercentHeld      | 91.81%   | 91.81%  | -0.00% |
| institutionsFloatPercentHeld | 92.04%   | 92.04%  | -0.00% |
| institutionsCount            | 1416.00  | 1404.00 | -12.00 |
| Months Compared              | 2026-06  | 2026-07 | -      |

Major Holders Snapshot

|                                               |                                          |
|-----------------------------------------------|------------------------------------------|
| <b>INSIDERSPERCENTHELD</b><br>0.25%           | <b>INSTITUTIONSPERCENTHELD</b><br>91.81% |
| <b>INSTITUTIONSFLOATPERCENTHELD</b><br>92.04% | <b>INSTITUTIONSCOUNT</b><br>1404.00      |

SEC Insider Transactions

|                            |                                  |
|----------------------------|----------------------------------|
| <b>AS OF</b><br>2026-07-01 | <b>BUY / SELL COUNT</b><br>1 / 2 |
|----------------------------|----------------------------------|

|                   |
|-------------------|
| <b>NET SHARES</b> |
| -1,394            |

|                  |
|------------------|
| <b>NET VALUE</b> |
| \$-521.43K       |

| Date       | Insider Name       | Label                                      | Direction | Shares | Price  | Owned After |
|------------|--------------------|--------------------------------------------|-----------|--------|--------|-------------|
| 2026-07-01 | Tomczyk Fredric J  | Payment Of Exercise Price Or Tax Liability | sell      | 337    | 247.72 | 35,631      |
| 2026-05-19 | Wilkinson Allen    | Grant Or Award                             | buy       | 166    | 0.00   | 166         |
| 2026-05-18 | Froetscher Janet P | Open Market Or Private Sale                | sell      | 1,223  | 358.09 | 13,807      |

Top Institutional Holders

| Holder                            | Shares | Value   | % Held | As Of      |
|-----------------------------------|--------|---------|--------|------------|
| Blackrock Inc.                    | 10.51M | \$2.55B | 9.42%  | 2026-03-31 |
| Vanguard Capital Management LLC   | 6.80M  | \$1.65B | 6.09%  | 2026-03-31 |
| Vanguard Portfolio Management LLC | 5.24M  | \$1.27B | 4.70%  | 2026-03-31 |
| Alliancebernstein L.P.            | 5.06M  | \$1.23B | 4.54%  | 2026-03-31 |
| State Street Corporation          | 4.98M  | \$1.21B | 4.46%  | 2026-03-31 |

Mutual Fund Holders

| Holder                                                      | Shares | % Out | Value     | As Of      |
|-------------------------------------------------------------|--------|-------|-----------|------------|
| VANGUARD INDEX FUNDS-Vanguard Total Stock Market Index Fund | 3.32M  | 2.97% | \$805.44M | 2026-03-31 |
| VANGUARD INDEX FUNDS-Vanguard 500 Index Fund                | 2.66M  | 2.38% | \$645.39M | 2026-03-31 |
| VANGUARD INDEX FUNDS-Vanguard Mid-Cap Index Fund            | 2.26M  | 2.03% | \$548.53M | 2026-03-31 |
| iShares Trust-iShares Core S&P 500 ETF                      | 1.35M  | 1.21% | \$328.46M | 2026-04-30 |
| Fidelity Concord Street Trust-Fidelity 500 Index Fund       | 1.34M  | 1.20% | \$324.97M | 2026-05-31 |

Insider Activity (yfinance)

| Filing Date | Insider Name           | Title                    | Trade Type   | Price  | Qty  | Value      |
|-------------|------------------------|--------------------------|--------------|--------|------|------------|
| 2026-05-18  | FROETSCHER JANET P     | Director                 | DOWN S       | 358.09 | 1223 | 437944.00  |
| 2026-02-23  | WILKINSON ALLEN        | Officer                  | DOWN S       | 292.79 | 248  | 72612.00   |
| 2026-02-19  | ISAACSON CHRISTOPHER A | Chief Operating Officer  | N/A Exercise | 286.17 | 5961 | 1705859.00 |
| 2026-02-19  | FOLEY STEPHANIE        | Officer                  | N/A Exercise | 286.17 | 1695 | 485058.00  |
| 2026-02-19  | WILKINSON ALLEN        | Officer                  | N/A Exercise | 286.17 | 787  | 225216.00  |
| 2026-02-19  | LIPSCOMB TIMOTHY       | Chief Technology Officer | N/A Exercise | 286.17 | 1882 | 538572.00  |
| 2026-02-19  | BHATIA PRASHANT        | Officer                  | N/A Exercise | 286.17 | 224  | 64102.00   |
| 2026-02-19  | HOCKING ROBERT A       | Officer                  | N/A Exercise | 286.17 | 355  | 101590.00  |

Insider Pressure Summary

|                         |                                  |
|-------------------------|----------------------------------|
| BUY COUNT<br>0          | SELL COUNT<br>2                  |
| NET VALUE<br>\$-510.56K | NET VALUE / MARKET CAP<br>-0.00% |



## Recent News

**NOTE**

Narrative: bullish bias in recent headlines.

Coverage: 6 relevant headlines reviewed; 6 dated items.

| Lens    | Headlines                                                                                                                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bullish | CBOE Q2 Earnings Beat on Options Strength, Revenue Outlook Raised ; CBOE Global (CBOE) Beats Q2 Earnings and Revenue Estimates                                                                                                                          |
| Bearish | No flagged headlines                                                                                                                                                                                                                                    |
| Neutral | Is Cboe Global Markets (CBOE) Stock Already Priced For Perfection? ; Cboe Global Markets Inc (CBOE) (Q2 2026) Earnings Call Highlights: Record Revenue and Strategic ... ; Wall Street Braces for Big Stock Moves That Could Roil Sleepy August Markets |

| Date       | Publisher       | Tag      | Impact          | Headline                                                                                            |
|------------|-----------------|----------|-----------------|-----------------------------------------------------------------------------------------------------|
| 2026-08-03 | Simply Wall St. | general  | [LOW] [RECENT]  | Is Cboe Global Markets (CBOE) Stock Already Priced For Perfection?                                  |
| 2026-07-31 | GuruFocus.com   | earnings | [HIGH] [RECENT] | Cboe Global Markets Inc (CBOE) (Q2 2026) Earnings Call Highlights: Record Revenue and Strategic ... |
| 2026-07-31 | Zacks           | earnings | [HIGH] [RECENT] | CBOE Q2 Earnings Beat on Options Strength, Revenue Outlook Raised                                   |
| 2026-07-31 | Barrons.com     | general  | [LOW] [RECENT]  | Wall Street Braces for Big Stock Moves That Could Roil Sleepy August Markets                        |
| 2026-07-31 | Zacks           | earnings | [HIGH] [RECENT] | CBOE (CBOE) Reports Q2 Earnings: What Key Metrics Have to Say                                       |
| 2026-07-31 | Zacks           | earnings | [HIGH] [RECENT] | CBOE Global (CBOE) Beats Q2 Earnings and Revenue Estimates                                          |

Short Sale Pressure

|                                     |                                      |
|-------------------------------------|--------------------------------------|
| AS OF<br>2026-07-31                 | SHORT VOLUME<br>478589.00            |
| SHORT EXEMPT VOLUME<br>11904.00     | TOTAL VOLUME<br>695472.00            |
| SHORT VOLUME RATIO<br>68.81%        | SHORT EXEMPT VOLUME RATIO<br>1.71%   |
| 5D AVG SHORT VOLUME RATIO<br>53.22% | 20D AVG SHORT VOLUME RATIO<br>51.71% |
| READ<br>Short pressure elevated     |                                      |

Short Interest Insight

| Lens        | Insight                                                          |
|-------------|------------------------------------------------------------------|
| Read        | short pressure elevated                                          |
| Implication | bearish pressure is rising in the tape                           |
| Watch Next  | watch short ratio versus 20D average and price response together |
| Confidence  | high                                                             |

Event Risk Calendar

| Event    | Date       | Distance | Note            |
|----------|------------|----------|-----------------|
| Earnings | 2026-10-30 | 87d      | scheduled       |
| Dividend | 2026-05-25 | 71d ago  | last cash event |
| 8-K      | 2026-07-31 | 4d ago   | latest filing   |

Catalyst / Event Risk

| Driver             | Status     |
|--------------------|------------|
| Next Earnings      | 2026-10-30 |
| Latest Dividend    | 2026-05-25 |
| Analyst Target Gap | 6.10%      |

SEC Filing Timeline

|                                  |                                  |
|----------------------------------|----------------------------------|
| LATEST FORM<br>8-K               | LATEST FILING DATE<br>2026-07-31 |
| LATEST REPORT DATE<br>2026-07-31 | LATEST FILING LAG<br>0.00d       |

8-K COUNT (90D)

4

10-Q COUNT (365D)

3

10-K COUNT (365D)

1

| Form | Filing Date | Report D... | Lag    | Type |
|------|-------------|-------------|--------|------|
| 8-K  | 2026-07-31  | 2026-07-31  | 0.00d  | 8-K  |
| 10-Q | 2026-07-31  | 2026-06-30  | 31.00d | 10-Q |
| 8-K  | 2026-07-28  | 2026-07-24  | 4.00d  | 8-K  |
| 8-K  | 2026-06-26  | 2026-06-23  | 3.00d  | 8-K  |
| 8-K  | 2026-05-18  | 2026-05-14  | 4.00d  | 8-K  |

End Summary

| Lens              | Close                             |
|-------------------|-----------------------------------|
| Profile           | full                              |
| Action Bias       | Hold / Neutral                    |
| Score / Verdict   | 52.45 / MONITOR                   |
| Risk Box          | stop 272.25 / trade target 348.99 |
| Analyst Mean Gap  | 6.10%                             |
| Flow Context      | \$-3.32M                          |
| Regime / Catalyst | neutral / 2026-10-30              |

Closing Insight

| Lens        | Insight                                                                        |
|-------------|--------------------------------------------------------------------------------|
| Read        | setup remains actionable into the next catalyst window                         |
| Implication | size to the setup only if flow, relative strength, and event path stay aligned |
| Watch Next  | watch stop discipline, benchmark-relative tape, and catalyst timing together   |
| Confidence  | medium / stale inputs                                                          |

Appendix

NOTE

Reference material, data provenance, and lower-priority diagnostics.

DATASET ROOT

cerberus\_data

FRESH / STALE / UNKNOWN

0 / 25 / 1

TRACKED DATASETS

26

| Dataset        | As Of      | Age     | SLA    | Status | Source         |
|----------------|------------|---------|--------|--------|----------------|
| Actions        | 2026-05-25 | 1715.6h | no SLA | stale  | cerberus_data  |
| SEC Insider    | 2026-07-01 | 831.6h  | no SLA | stale  | cerberus_data  |
| NAAIM          | 2026-07-22 | 327.6h  | no SLA | stale  | cerberus_data  |
| aaii_sentiment | 2026-07-23 | 303.6h  | no SLA | stale  | aaii_sentiment |
| cboe_bxm       | 2026-07-31 | 111.6h  | no SLA | stale  | cboe_bxm       |
| cboe_longvol   | 2026-07-31 | 111.6h  | no SLA | stale  | cboe_longvol   |
| cboe_lovol     | 2026-07-31 | 111.6h  | no SLA | stale  | cboe_lovol     |
| cboe_ltv       | 2026-07-31 | 111.6h  | no SLA | stale  | cboe_ltv       |

+18 additional datasets tracked in companion manifest.

Methodology

NOTE

#### Definitions

| Topic                 | Note                                                                                                                                                                                                                                         |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score Components      | Momentum, Trend, Mean Reversion, Bollinger, VWAP, Trend Strength, Vol Stability, Risk Adjusted, Drawdown, Volume Confirmation                                                                                                                |
| Flow Divergence Logic | Flow divergence compares 5D/20D price returns against cumulative major and retail net flow from local stock_flows; positive major flow with flat/down price flags accumulation, negative major flow with resilient price flags distribution. |
| Benchmark Selection   | Benchmarks prefer index membership and sector mapping from local security/index snapshots; current order SPY, XLF.                                                                                                                           |

#### Fallback Usage

| Dataset             | Handling                                | Severity |
|---------------------|-----------------------------------------|----------|
| fundamentals        | fundamentals dataset unavailable        | optional |
| options_summary     | options_summary dataset unavailable     | optional |
| sec_company_tickers | sec_company_tickers dataset unavailable | optional |

NOTE

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## Document Control

|                |                                    |
|----------------|------------------------------------|
| Report Schema  | 2                                  |
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